Role-Differentiated Learning within Organizations

Abstract

Research Summary: Existing research has yet to reveal how learning is divided among members within organizational units. We develop a role-differentiated framework of learning in production teams, in which workers and managers develop distinct learning trajectories. Using detailed task-level records covering 79,639 products at a Chinese electrical-equipment manufacturer, we test the predictions of our framework. We find that workers' experience significantly improves execution performance, whereas managers' experience significantly improves coordination performance. Managers' learning in coordination becomes stronger when they actively coordinate production by reordering tasks and reallocating workers. Worker and manager experience are also complementary, as accumulated worker experience strengthens managers' ability to translate their own experience into shorter waiting time.

Managerial Summary: Firms often track experience through individual tenure or total team output. Our findings suggest that firms should first ask where experience resides and which operating problem it can solve. Frontline workers' experience accumulated through repeated execution may be of limited value when the bottleneck lies in hand-offs and coordination, where managers' experience is more important. Digital production and human-resource systems can support such decisions by making role-specific learning trajectories visible and tracking shared experience within particular role combinations and products. Firms can also actively shape how experience develops by encouraging employees to engage in activities central to their roles. At the same time, experience accumulated in one role can increase the returns to experience in another. Firms should therefore manage experience as a relational rather than purely individual resource, considering how experience is combined across interdependent roles in team formation and assignment.

Keywords: Organizational learning; Learning-by-doing; Middle managers; Coordination

1 INTRODUCTION

Organizational learning is a fundamental source of productivity improvement and competitive advantage (Hatch & Dyer, 2004; Teece, 2007). Organizations do not accumulate the experience from which they learn as unitary actors; rather, they do so through members whose distinct roles expose them to different tasks and problems (Argote & Miron-Spektor, 2011; Bechky, 2003; Simon, 1991). Understanding where this learning resides is therefore important both theoretically and practically. Theoretically, an aggregate view of organizational learning can obscure the mechanisms through which accumulated experience produces performance improvements (Maula et al., 2023). Practically, this ambiguity complicates decisions about team composition, retention, rotation, and the deployment of experienced personnel. Organizations may therefore act sub-optimally by preserving experience that does little to improve the performance dimension that constrains them or by breaking up combinations whose shared experience could address it.

Yet how learning is distributed within organizational units remains largely unexamined. Early empirical studies inferred learning at the firm or team level from the relationship between accumulated experience and overall performance, but could not reveal finer-grained patterns—for example, whether members within an organization learn at comparable rates, acquire the same knowledge, or improve the same outcomes (e.g., Schilling et al., 2003; Taylor & Greve, 2006). Subsequent studies distinguished among individual-, team-, and firm-level experience, but they neither differentiated it by members' roles within the same unit nor examined learning across multiple performance dimensions (Fong Boh et al., 2007; Jain, 2013; Reagans et al., 2005). These omissions may conceal which performance dimensions improve with experience accumulated in different roles and how these role-specific learning trajectories depend on one another.

To address this gap, we develop a role-differentiated account of learning within production teams and test its predictions. A role is a characteristic pattern of behavior associated with a particular organizational position (Biddle, 1986). Production teams have a simple

structure comprising two roles: team managers and frontline workers. These roles differ in the responsibilities they entail, the actions they involve, and the feedback their occupants receive, so experience accumulated in each may improve different dimensions of team performance. Moreover, these role-specific learning trajectories may complement one another and jointly improve overall organizational performance.

We test these predictions using unusually detailed task-level records covering 79,639 products across 3,556 models at a Chinese electrical-equipment manufacturer. For each product, the records reconstruct the complete production sequence, identify the frontline worker and corresponding team manager responsible for each task, and record outcomes related to production efficiency, quality, waiting times between interdependent tasks, and material availability. These outcomes allow us to construct distinct measures of execution and coordination performance, thereby extending the empirical study of organizational learning to coordination performance.[1] Because the same models are produced by different combinations of workers and managers over time, we can distinguish the model-specific shared experience accumulated within each role from general learning associated with cumulative production. Together, the data's granularity, repeated observations, and role-specific measures allow us to test whether workers' and managers' experience improves execution or coordination and whether their learning trajectories complement one another—relationships that organization- or team-level data typically cannot disentangle.

The results reveal a division of learning that aggregate team learning curves would obscure. Workers' shared experience primarily improves execution, whereas managers' shared experience primarily improves coordination. Doubling worker-combination experience reduces the probability of a defect by 0.7 percentage points, or 6.4 percent relative to the mean. Doubling manager-combination experience reduces waiting time between tasks by 2.7 percent and the probability of material shortages by 0.59 percentage points. Managers' learning is stronger when they enact their coordinating role by reordering tasks and real-

[1] See Table A1 for dependent variables used in empirical studies on organizational learning, and Table A19 and Figure A17 in Appendix H for a review.

locating workers. The two forms of experience also complement one another, particularly in improving coordination: workers' accumulated experience increases managers' ability to translate their own experience into smoother production flow. Together, the framework and findings show that organizational learning depends not simply on how much experience a team possesses, but on who accumulates it, which performance problem it can solve, and how managerial action enables team members to develop and integrate specialized knowledge across roles.

This study contributes to research on organizational learning in four ways. First, we locate learning within hierarchical teams by distinguishing workers' shared experience from team managers' shared experience and showing that the two improve different dimensions of performance. This distinction shows that learning is distributed across hierarchical roles rather than embedded in teams as an undifferentiated stock of shared experience. Second, we examine multiple theoretically distinct outcomes within the same organizational setting. Prior studies often examine a single performance dimension, making it difficult to determine whether divergent findings reflect distinct learning mechanisms or variation across empirical settings. By holding the broader production technology, institutional environment, and measurement system constant, we show that learning is outcome-specific. Third, we identify cross-hierarchy complementarities in learning. Prior studies consider complementarities between experience measured at different organizational levels (Espinosa et al., 2007; Jain, 2013) or accumulated during different periods (Kor, 2003; Zheng et al., 2016). We extend this work by showing that learning trajectories across interdependent roles within the same team can also be complementary. Prior research focuses mainly on how managers affect workers; we document the less-studied reverse relationship: workers' accumulated experience strengthens team managers' ability to translate their own experience into improved coordination performance. Fourth, we show that managerial actions and skills shape how role-specific learning is generated and integrated. Prior theoretical work argues that managers influence organizational learning through learning routines, cognitive capability and learning-related

resources (Harvey et al., 2022; Helfat & Martin, 2015; Helfat & Peteraf, 2015; Hodgkinson & Healey, 2011), but empirical evidence remains scarce. We show that managers' coordination interventions strengthen the effect of their experience on coordination performance. Together, these contributions advance our understanding of organizational learning by showing not only that organizational members may learn different things, but also that formal roles, hierarchical relationships, and managerial behavior shape where learning resides and how it contributes to the development of dynamic capabilities.

We also contribute to research on managers and dynamic capabilities. Recent studies show that managers shape worker productivity through allocation, monitoring, coaching, and people-management skills, but largely treat managerial capability as a relatively stable attribute (Adhvaryu et al., 2022; Benson & Shaw, 2025; Diaz et al., 2025; Hoffman & Tadelis, 2021; Lazear et al., 2015). Our findings add a dynamic perspective by showing how managers develop coordination capability through shared experience and how workers' learning accelerates this process.

Our study is also closely related to Adhvaryu et al. (2023), who show that supervisors' non-cognitive skills accelerate worker learning in garment-manufacturing lines. Our study differs from theirs in three aspects: we examine managerial experience as an organizational resource that accumulates over time rather than managerial quality as a relatively stable attribute; distinguish the effects of workers' and managers' learning on execution and coordination outcomes; and show that workers' accumulated experience can, in turn, strengthen the coordination gains from managers' experience.

2 THEORY AND HYPOTHESES

Organizational learning is defined as a process through which organizations' experience performing a task is converted into knowledge, which, in turn, changes the organization and affects its future performance (Argote et al., 2021). Here, the learning subject can be a

firm, plant, or team, depending on where experience accumulates and routines are enacted. Critically, organizational learning is not reducible to the aggregation of individual learning because it relies on supra-individual repositories, such as routines, role structures, and relational ties, and on collective processes that persist beyond any single member (Espinosa et al., 2007; Fong Boh et al., 2007; Huckman et al., 2009; Jain, 2013; Reagans et al., 2005; Taylor & Greve, 2006).

Given the importance of organizational learning and the substantial variation in firms' learning capabilities (Balasubramanian & Lieberman, 2008; Teece, 2007), scholars have called for closer attention to the mechanisms through which organizations learn (Argote & Ingram, 2000; Spender & Grant, 1996). Argote et al. (2021) emphasize that organizational learning occurs within a division of labor in which members perform different tasks and must learn to coordinate their interdependent activities. Yet existing theory has not explained how the division of labor gives rise to a corresponding division of learning. It therefore remains unclear how this division of learning develops, what factors shape it, and how it influences learning outcomes at the organizational-unit level.

We address this gap with a role-differentiated learning framework. Roles are sets of expected behaviors associated with particular positions (Ilgen & Hollenbeck, 1991; Katz et al., 1978). We develop our framework in the setting of production teams, where members occupy one of two clearly differentiated roles: frontline workers or team managers. Roles systematically structure members' responsibilities, actions, and feedback. As a result, members in different roles receive different information and develop familiarity with different aspects of production through learning by doing, giving rise to distinct learning trajectories. Because members occupying the same role often work together to perform a particular type of work, our framework focuses on how the shared experience accumulated by specific combinations of members within each role contributes to improvements in team performance. In what follows, we explain how learning within each role contributes to different dimensions of team performance and how the resulting learning trajectories relate to one another.

2.1 Role-differentiated Learning

We distinguish two dimensions of production performance. Execution performance captures the speed, accuracy, and reliability with which focal tasks are completed. Examples include task completion time and output quality in manufacturing, procedure time in hospitals (Pisano et al., 2001; Ramdas et al., 2018; Reagans et al., 2005), and project completion time in software development (Espinosa et al., 2007; Narayanan et al., 2009). Coordination performance captures how effectively interdependent tasks are integrated through handoffs, interactions, and the alignment of information and resources. Examples include inter-task waiting time and material-supply efficiency in manufacturing; patient routing, emergency-department waiting time (Dong et al., 2019), and cross-role communication within care-provider teams (Hoffer Gittell, 2002); and the effectiveness of handoffs and schedule synchronization in airline operations.This distinction matters because an aggregate productivity gain can result from better task execution, better coordination among tasks, or both.

Role specialization does not make differentiated learning automatic. People can learn vicariously by observing and interpreting others' experience (Bandura & Walters, 1977; Myers, 2018). Workers may observe how managers resolve shortages or resequence work, while managers may learn about task execution by observing workers. Moreover, because execution and coordination are interdependent, experience held by one role can affect the performance of the other. These possibilities yield a plausible alternative to role differentiation: both roles could learn about both performance dimensions. Our prediction is therefore comparative rather than exclusive. Roles do not prevent learning across domains; they concentrate direct problem exposure, action, feedback, and accountability within domains.

Workers repeatedly handle materials, operate equipment, follow technical procedures, and correct immediate errors (Pisano et al., 2001; Reagans et al., 2005; Thurstone, 1919). They receive direct feedback from defects, rework, and the outcomes of completed tasks. Shared experience allows a worker combination to recognize recurring mistakes, refine movements and procedures, and stabilize effective routines. These learning opportunities map most

directly onto execution. They provide less opportunity to learn about decisions that workers neither control nor observe completely, for example, how to sequence interdependent tasks, allocate labor across tasks, or secure inputs across teams.

Managers repeatedly assign workers, sequence tasks, manage handoffs, monitor progress, and respond to bottlenecks and exceptions (Adhvaryu et al., 2022; Benson & Shaw, 2025; Lazear et al., 2015; Sirmon et al., 2007, 2011). They receive feedback from workflow-level outcomes such as delays, idle time, missed handoffs, disruption patterns, and resource constraints. Shared experience allows a manager combination to recognize recurring dependencies, anticipate disruptions, and refine routines for allocation, escalation, and exception handling. These learning opportunities map most directly onto coordination. Managers may also influence execution, but they do not directly repeat the focal production tasks through which task-specific execution routines develop.

The firm's incentives reinforce this role-based concentration of learning. Workers are directly accountable for traceable quality problems, while individual speed has less direct influence on their compensation and is harder to attribute to a single worker. Workers should therefore devote more attention to learning how to avoid defects than to shortening operating time. Managers are accountable for delivering defect-free products on time and control the allocation, sequencing, and resource responses that shape production flow. Their incentive to improve speed and reliability therefore operates primarily through coordination. More generally, experience should generate the largest improvements on dimensions for which a role combines direct responsibility, discretion, diagnostic feedback, and accountability (Baker, 1992; Feltham & Xie, 1994; Holmstrom & Milgrom, 1991; Kerr, 1975).

Hypothesis 1. *Workers' shared experience has a stronger positive effect on execution performance than on coordination performance, whereas managers' shared experience has a stronger positive effect on coordination performance than on execution performance.*

2.2 Managerial Coordination Interventions and Learning

Experience creates an opportunity to learn, but the information obtained from experience depends on what members do. Managers can respond passively to recurring production conditions, or they can intervene by changing task sequences and reallocating workers across tasks. These interventions are consequential experiments: each adjustment changes a concrete worker-task configuration and reveals its effects on handoffs, bottlenecks, and resource constraints. Repeated intervention therefore produces more diagnostic feedback than repeated exposure alone (Arrow, 1962; Kolb, 2014). It helps managers update causal representations of the workflow, for instance, which dependencies are fragile, which assignments work under pressure, and which responses resolve recurring disruptions, and convert those representations into more effective coordination routines (Camuffo et al., 2020, 2024; Frese & Gielnik, 2023). Because managers have authority over sequencing and allocation, they can act on this feedback; workers assigned to focal tasks generally cannot.

Managerial intervention can also change the quality of workers' experience. Stable access to inputs, clearer handoffs, and faster resolution of disruptions make repeated execution less fragmented and its feedback easier to interpret. Under these conditions, workers can identify the source of an error, compare successive attempts, and retain effective task routines. Task reordering and worker reallocation can therefore strengthen not only the return to managers' coordination experience but also the return to workers' execution experience. In contrast, reassignment may also curtail the repetition through which task-specific routines become reliable. We nevertheless expect the informational benefit of intervention to dominate in our setting. Customized production repeatedly creates new bottlenecks and assignment mismatches, and managers can observe the consequences of the sequencing and allocation changes they initiate. Problem-responsive intervention should therefore generate more useful feedback than passive repetition of an imperfect workflow.

Hypothesis 2. *More active managerial intervention in coordination activities will (i) increase the rate at which managers improve coordination performance and (ii) increase the*

rate at which workers improve execution performance.

2.3 Cross-role Complementarity

Organizations' collective value can arise from complementarities among differentiated human capital rather than from isolated individual contributions (Crocker & Eckardt, 2014; Espinosa et al., 2026; Han & Won, 2026; Kryscynski et al., 2021; Ployhart & Moliterno, 2011; Wolfson & Mathieu, 2018). Yet complementarity is not guaranteed: experienced managers might substitute for inexperienced workers, or experienced workers might reduce the need for managerial coordination. Complementarity arises only when the knowledge accumulated in one role increases the performance value of knowledge accumulated in the other. In hierarchical production, this condition is most likely when an outcome depends jointly on task execution and production flow.

For execution, experienced workers can perform faster only when materials arrive, handoffs occur on time, priorities are clear, and exceptions are resolved. Experienced managers learn to create these conditions by anticipating bottlenecks, sequencing tasks, and reallocating labor (Adhvaryu et al., 2022). Manager experience should therefore increase the extent to which worker experience improves speed, throughput, or cycle time. The same amplification is less likely for quality outcomes. Quality problems arise more directly from workers' task routines and attention to technical requirements, leaving less scope for managers' coordination experience to raise the return to worker experience.

For coordination, experienced managers can improve the workflow more effectively when task durations and handoffs are sufficiently reliable to diagnose the source of a delay. Experienced workers make task completion more stable and predictable, giving managers clearer feedback about which coordination routines work. Worker experience should therefore increase the extent to which manager experience improves inter-task handoffs. The same amplification is less likely for input management. Forecasting and securing materials lie primarily within the managerial domain, so worker experience contributes less directly to this

outcome.

Hypothesis 3. *Worker experience and manager experience are complementary: worker experience generates greater improvements in execution performance when manager experience is higher, and manager experience generates greater improvements in coordination performance when worker experience is higher.*

2.4 Managerial Soft Skills Strengthen Complementarity

Cross-role complementarity requires more than the coexistence of execution and coordination knowledge. Workers must communicate local problems, managers must interpret that information, and both roles must cooperate when assignments or task sequences change. Managerial soft skills, the ability to communicate clearly, understand workers' constraints, build trust, and adapt one's interpersonal approach, can improve each step of this interaction. Social skills reduce coordination costs in team production (Deming, 2017, 2022; Heckman & Kautz, 2012; Hoffman & Tadelis, 2021), while high-quality manager-worker relationships facilitate information sharing, trust, and cooperation (Dienesch & Liden, 1986; Graen & Uhl-Bien, 1995; Liden et al., 1997; Wayne et al., 1997). Managers with stronger soft skills should therefore receive earlier and more accurate information about execution problems and obtain greater cooperation when they adjust the workflow.

This moderating role is clearest for coordination outcomes because diagnosing delays and changing handoffs depend directly on communication and joint adjustment across roles. When managers have stronger soft skills, worker experience should provide a more useful basis for managers to apply their coordination experience. For execution outcomes, two forces compete. Stronger soft skills may help managers remove disruptions and give workers clearer feedback, strengthening the complementarity between worker and manager experience. But responsive managers may also solve problems on workers' behalf, reducing workers' need to develop independent execution routines. The net moderating effect on execution is therefore indeterminate.

Hypothesis 4. *Managerial soft skills strengthen the complementarity between worker experience and manager experience for coordination outcomes. For execution outcomes, the direction of this moderating effect is theoretically indeterminate.*

Together, H1–H4 constitute our role-differentiated learning framework, which is summarized in Figure 1.

INSERT FIGURE 1 HERE

3 METHODS

This section describes the data and empirical strategy used to test our hypotheses. We begin by introducing the research setting and the organization of production, which describe how workers and managers interact within teams and face different incentives (Section 3.1). We then describe the data sources, motivate our unit of analysis at the product level, and construct our key variables, with a particular focus on measuring shared experience at the level of worker and manager combinations (Sections 3.2 and 3.3). Next, we outline our identification strategy, which exploits variation in cumulative experience across product sequences in an environment with frequent model changes (Section 3.4). Finally, we present our empirical specifications, linking worker and manager experience, as well as their interaction, to different dimensions of performance, including execution outcomes (operating time and defects) and coordination outcomes (waiting time and material shortages) (Section 3.5).

3.1 Research Setting

We partner with a Chinese electrical-equipment manufacturer founded about 20 years ago. The firm operates a team-based production system organized around 38 production teams. Within each team, a team manager leads frontline workers in production. This structure provides a useful setting for studying role-differentiated learning because workers and managers contribute to the same products while occupying distinct organizational roles.

Models and Products The firm uses a customized make-to-order production system. After a customer signs a contract, the engineering department finalizes the design specifications. These specifications define a distinct product model. Models vary substantially in size, complexity, and technical requirements. A model may be manufactured multiple times, with each production instance corresponding to an individual product. Each product, in turn, consists of an ordered sequence of production tasks. Thus, tasks are nested within products, and products are nested within models.

For example, a customer may order 100 electrical cabinets built to the same design specifications. These specifications define one model, while each of the 100 cabinets constitutes a separate product. Once the model design is finalized, senior management assigns the products in the order to production teams based primarily on workload and task characteristics rather than past team performance. Producing an individual cabinet requires an ordered sequence of tasks, such as welding the frame, installing electrical components, wiring, testing, and final inspection. Each product is typically handled by two or three teams, with each team responsible for a different part of the task sequence. The set of workers and managers in the multiple teams who work on the same product, rather than the members of a single team, forms the analysis unit in our empirical study.

Responsibilities, Actions, and Feedback The production process creates a clear division of responsibility between workers and managers. Workers are primarily responsible for executing assigned tasks accurately. Their main actions include following task-specific technical requirements, operating equipment, handling materials, completing procedural steps, and correcting errors. Workers receive feedback through repeated production, observation of their own work outcomes, defect reports, and guidance from team managers.

Team managers are primarily responsible for coordinating production and ensuring timely completion. They allocate tasks, arrange schedules, monitor progress and handoffs, and adjust production when problems arise or conditions change. Managers receive feedback

from workflow outcomes, including delays, bottlenecks, idle time, missed handoffs, rework, defects, and delivery performance.

Incentive Structures and the Quality–Efficiency Trade-off The firm's incentive system is designed to balance quality and efficiency, but workers and managers face different incentive structures. Workers' incentives emphasize quality more strongly than efficiency, for two reasons. First, in how pay is determined: as the HR manager explained, "*Task speed is digitally recorded and visible to managers, but it is not directly used to determine individual compensation. Workers' bonuses are primarily tied to team-level output and to a performance score assigned by the team manager. Within this score, product quality plays a central role, whereas speed and punctuality are secondary.*" Second, in how accountability works in practice: quality violations are digitally traceable and followed up, so workers bear the consequences of their own mistakes, whereas the overall pace of a product is hard to attribute to any single worker.

Managers, by contrast, are evaluated on both quality and efficiency, because they are accountable for the final outcome of the product delivered to the customer, which must be both defect-free and on time. They are responsible for preventing defects and rework, which themselves delay completion, and they are also accountable for workflow speed and timely delivery.

3.2 Data

Production Data Our primary data source is the firm's digital production tracking system. From January 2022 to June 2024, the firm produced 79,639 individual products across 3,556 distinct models. The number of products produced per model varies substantially: the most popular model includes 319 products, while the median model includes six. Figure A1 presents this distribution, and Appendix A.2 provides a detailed illustration of the data structure. For each product, the tracking system records every production task. Each

task record identifies the assigned worker, start and end times, and whether a quality issue or material shortage occurred. [2]

Administrative Data Our second data source is the firm's human resource management records, covering the period from January 2022 to June 2024. This dataset provides detailed information on each worker's position, team assignment, and team manager, as well as demographic characteristics including gender, age, hiring date, education, and ethnicity. Linked to the production records, these data identify each worker's team and manager and allow us to observe task allocation within teams.

Survey Data In September 2024, we administered an online survey to all front-line workers, which took an average of 15 minutes per worker. Among 299 workers who appear in the production data, 156 were active as of September 2024, all of whom completed the survey. A potential concern is selective attrition: workers who remained with the firm may have had better relationships with their team managers, which could bias our survey-based measure of managerial soft skills. Appendix A.3 mitigates this concern by showing that attrition is unrelated to observable worker characteristics or prior team performance. The questionnaire included items evaluating workers' perceptions of how effectively their managers engaged with them, covering responsiveness, problem diagnosis, problem-solving, understanding of workers, and communication. Although the survey was not anonymous, respondents were assured of the confidentiality of their answers. We use the responses of the 156 workers to construct our measure of managerial soft skills, which allows us to examine whether cross-role complementarity varies with perceived managerial soft skills.

[2]The production data include 299 unique workers. The firm has roughly 800 active employees, but many work in upstream or support departments, such as material preparation, whose activities are not captured in the production-stage tracking system. Within the production data used in our analysis, worker assignments are fully observed.

3.3 Variables

3.3.1 Dependent Variables

The digital production records allow us to construct measures that capture the role-based distinction between execution and coordination performance developed in Section 2.1. For execution, we start from the smallest production unit we observe: a task performed by an individual worker. Workers directly control how quickly and accurately the task is completed, allowing us to construct measures of task speed and quality. For coordination, we focus on outcomes that depend on the coordination of interdependent tasks and resources. Because we use successive products of the same model to identify learning of workers and managers, we aggregate these task-level records to construct performance outcomes at the product level.

Execution Performance We first consider the efficiency of task execution. We construct operating time as the total duration of all tasks within a product, calculated as the sum of individual task durations for that product. This measure isolates the portion of production time attributable to task execution, which is thus directly under the control of the workers performing those tasks. Operating time is a standard measure of production efficiency in studies of organizational production performance (Espinosa et al., 2007; Pisano et al., 2001; Reagans et al., 2005). The second measure captures the quality of task execution, which is widely recognized as a key dimension of production outcomes (Eggers, 2012; Hatch & Dyer, 2004; Levitt et al., 2013). We proxy quality with an indicator for a defect, which equals one if any task within a product was re-executed due to quality issues.

Coordination Performance We first construct inter-task waiting time as total production duration minus the sum of individual task durations, excluding non-working hours for cross-day production (see Appendix A.4 for details on the variable construction). This measure captures the efficiency of handoffs between consecutive production tasks. This construction follows prior operations-management studies of coordination efficiency (Batt &

Terwiesch, 2017; Dong et al., 2019), but its implications for organizational learning have received little attention. Second, we construct an indicator for material shortage, which equals one if any task within a product was flagged for material unavailability during production.

Daily Output In addition, we use daily output volume as a measure of overall productivity to validate that learning on execution and coordination outcomes translates into aggregate productivity gains. Because learning is identified within models, we construct this output measure at the model–day level. We define daily output as the number of products of a given model completed on that day. Across 39,541 model–day observations, average daily output is 2.75 products, with zero products completed in 58.9% of observations.

3.3.2 Independent Variables

To capture the learning processes of different organizational roles, we use shared experience within manager and worker combinations as our key independent variables. We define a manager (worker) combination as the exact set of managers (workers) who jointly work on the same product. For each model, the shared experience of managers (workers) is measured as the cumulative number of products completed by that same manager (worker) combination. These measures reflect the extent to which a given set of managers or workers has accumulated familiarity through prior collaboration.

By construction, shared experience increases by one with each additional product completed by the same combination; a new combination begins with zero shared experience for that model.[3] Manager and worker combinations can change within the production of

[3]For example, suppose managers A and B jointly produce the same model across ten successive products, with manager combination {A,B} staying intact throughout. Its shared experience therefore accumulates steadily from 0 (before the first product) to 9 (before the tenth). The worker combination, by contrast, changes over this sequence. Products 1–3 are produced by worker combination {1,2}, products 4–6 by a different combination {1,3}, and products 7–10 by {1,2} again. Each combination's shared experience counts only its own prior products of this model, so {1,2} accumulates experience 0, 1, and 2 over products 1–3, and the newly formed {1,3} starts afresh at 0, 1, and 2 over products 4–6. Crucially, when {1,2} returns at product 7, its experience is 3 rather than 0, because it had already completed three products (products 1–3), and it then continues to 4, 5, and 6 over products 8–10. Worker shared experience across the ten products is thus 0, 1, 2, 0, 1, 2, 3, 4, 5, and 6, respectively. A graphical illustration of this example, together with an illustration based on the actual data, is provided in Appendix A.5.

a model, on average 7.6 times for workers and 4.6 times for managers. These assignment changes across products reflect operational needs. As the head of production explained, "*We often reshuffle teams across products. This is just a routine driven primarily by scheduling and capacity needs rather than by team or worker performance.*" Because production is carried out collectively, we include worker and manager shared experience simultaneously. Aggregate team experience cannot distinguish learning between the two roles, while only observing the experience of a single role alone would conflate its learning with that of the other as their experience evolves together.

3.3.3 Measures of Managerial Interventions

Since our task-level production data record the worker assigned to each task and the order in which tasks are performed, we can observe whether managers actively coordinate the production process. We therefore construct two measures of managerial intervention: task reordering captures whether managers change the sequence of tasks across consecutive products, and worker reallocation captures whether managers reassign workers to tasks.

Task reordering is defined as a change in the sequence of production tasks across consecutive products within the same model. Worker reallocation is defined as a change in worker-to-task assignments across consecutive products within the same model. For each model, we construct the corresponding rate as the share of products exhibiting such changes; construction details and examples appear in Appendix A.6. 52% of models exhibit no task reordering, and conditional on any reordering, the average rate is 44%. Similarly, 51% exhibit no worker reallocation, and conditional on any reallocation, the average rate is 43%. Although the two measures capture distinct managerial decisions, namely changes in task order and changes in worker assignments, they are correlated (Appendix Figure A5), consistent with both reflecting managers' active adjustment of production arrangements.

3.3.4 Measures of Managerial Soft Skills

To capture managers' capacity for cross-role interaction, we construct a composite index of managerial soft skills using worker survey evaluations across five dimensions: responsiveness, problem diagnosis, problem-solving, understanding of workers, and communication. We aggregate these measures to the model level by taking the average score across workers involved in producing the same model. We then summarize the five model-level measures into a single index using principal component analysis (PCA) and classify models as high or low in managerial soft skills depending on whether their PCA score is above or below the sample median. As a robustness check on this PCA-based measure, we also construct this index using factor analysis. The two approaches yield nearly identical indices, with the resulting measures highly correlated and loading positively on all five dimensions, with details provided in Appendix A.7.

3.3.5 Summary Statistics

Table A3 presents summary statistics for the main variables. At the model level, the dataset includes 3,556 models, with an average of 22.40 products per model. Worker combinations change an average of 7.56 times within a model, compared with 4.58 changes in manager combinations. Among models for which managerial interventions can be measured, task reordering and worker reallocation occur at average rates of 21.1% and 21.2%, respectively. The product-level dataset includes 79,639 products. The average operating time is 6.84 hours, whereas the average waiting time is 26.26 hours, suggesting that idle periods between tasks consume substantially more time than task execution itself and may be an important source of production inefficiency. About 10.8% of products experienced at least one defect, and 65.4% encountered a material shortage. Worker combinations completed an average of 5.95 prior products (s.d. 11.87), while manager combinations completed an average of 16.40 prior products (s.d. 25.47), reflecting the wide dispersion in accumulated shared experience across products.

3.4 Empirical Strategy

Our identification combines frequent introduction of new models and changes in team composition within model sequences. To identify learning effects, we build on Levitt et al. (2013), who exploit the introduction of new product models as a source of plausibly exogenous variation in production experience. Because our partner firm predominantly engages in customized production, it frequently introduces new product models to meet customer-specific requirements, creating a continual need for production teams to learn. Specifically, each new model marks the starting point of a learning curve, while worker and manager combinations turn over independently within the model, allowing us to separately trace the experience accumulated in each role. This separation enables us to identify whether the performance benefits of learning differ across roles. Within a model, the worker combination changes twice and the manager combination changes once, at the median.

A key challenge to our identification strategy is that senior management may allocate new product models across teams based on their past performance. Because model-specific experience starts at a low level when a team begins producing a new model, performance-based assignment would make observed experience systematically related to expected team performance and thereby bias the estimated learning effects. For example, if stronger teams are more likely to receive new models, they may exhibit better performance despite having lower model-specific experience, biasing the estimated learning effects downward.

To address this concern, we test whether the allocation of new product models across teams is systematically related to their past performance. In Table A4, current-month assignment outcomes (the number of new models, output per model, and workload) are regressed on lagged performance (workload, output, and manager bonuses) and team characteristics (team size and the shares of female, Han-ethnicity, and college-educated workers). The coefficients are generally small and insignificant; team size predicts the number of new models assigned, but as a stable team characteristic it is absorbed by the team fixed effects in our main specification. The joint F-statistics (2.51, 0.51, and 0.92) and low R^2 confirm that past

performance has little power to predict future assignments, suggesting that our identification of learning is unlikely to suffer from the bias described above.

We now turn to identifying role-specific curves, which requires that the placement of individuals into combinations not be driven by their past performance, so that each role's experience remains exogenous. Table A5 regresses the same three assignment indicators on each worker's lagged average workload per model, lagged monthly wage, and cumulative experience. None significantly predicts assignment; the joint F-statistics (0.23, 1.21, 0.32) and low R^2 values indicate that individual workers are not allocated to models on the basis of past performance.

Selection could also operate through persistence rather than assignment: even if models are allocated at random, better-performing teams might be kept together for longer spells, mechanically accumulating more experience and again confounding learning with selection. We test this by regressing the total number of models a team completes over the sample period on its early performance—the average output and defect rate across its first two (or three) completed models. As reported in Table A6, early performance does not significantly predict a team's total number of models, indicating that teams with stronger initial performance are not retained for longer sequences. Together, these tests suggest that experience accumulation is unlikely to be driven by selection on past performance, whether through assignment or retention.

3.5 Specifications

Baseline Specification We estimate learning effects by relating cumulative production experience to product-level performance, following standard learning-curve approaches (Benkard, 2000; Egelman et al., 2017; Epple et al., 1996; Hatch & Dyer, 2004; Levitt et al., 2013). To test H1, our main specification includes separate measures of worker-combination and manager-combination experience as explanatory variables, which allows us to identify which

dimensions of performance improve with experience accumulated in each role:

$$Y_{jmt} = \alpha + \beta_1 \cdot \text{IHS}(\text{WorkerExp}_{w(jmt)}) + \beta_2 \cdot \text{IHS}(\text{ManagerExp}_{g(jmt)}) + \gamma_m + \delta_t + \theta_w + \lambda_g + \varepsilon_{jmt} \quad (1)$$

where the dependent variable Y_{jmt} captures product-level production outcomes for product j of model m in month t. The key explanatory variables are the cumulative numbers of same-model products previously completed by the respective worker and manager combinations before the production of product j. The experience variables are transformed using the inverse hyperbolic sine (IHS) function.[4] We include model fixed effects, γ_m, and month fixed effects, δ_t. These absorb differences in task complexity across models and time-specific factors such as order volume and climate conditions. We also include worker- and manager-combination fixed effects, θ_w and λ_g, to account for baseline productivity differences across combinations. These fixed effects help isolate within-combination changes in performance as experience accumulates. Standard errors are clustered at the model level to account for potential correlation in unobserved shocks or production conditions across products of the same model, and this clustering approach is applied consistently across all regression specifications.

How Coordination Interventions Shape Learning To test H2, which predicts that managers' coordination interventions strengthen both learning trajectories, we estimate separate specifications for the two dimensions of performance. For execution outcomes, we estimate:

$$\begin{aligned} Y_{jmt} = \alpha &+ \beta_1 \cdot \text{IHS}(\text{WorkerExp}_{w(jmt)}) + \beta_2 \cdot \text{IHS}(\text{ManagerExp}_{g(jmt)}) \\ &+ \beta_3 \cdot \text{IHS}(\text{WorkerExp}_{w(jmt)}) \times I_m + \gamma_m + \delta_t + \theta_w + \lambda_g + \varepsilon_{jmt} \end{aligned} \quad (2)$$

[4]We apply the inverse hyperbolic sine (IHS) transformation to operating time and waiting time, both of which measure time flows through the production process, with waiting time also taking negative values. The IHS accommodates zero and negative values while preserving a log-like interpretation.

For coordination outcomes, we estimate:

$$Y_{jmt} = \alpha + \beta_1 \cdot \text{IHS}(\text{WorkerExp}_{w(jmt)}) + \beta_2 \cdot \text{IHS}(\text{ManagerExp}_{g(jmt)}) + \beta_3 \cdot \text{IHS}(\text{ManagerExp}_{g(jmt)}) \times I_m + \gamma_m + \delta_t + \theta_w + \lambda_g + \varepsilon_{jmt} \quad (3)$$

where I_m denotes one of two measures of the intensity of managerial coordination interventions for product model m. The first is the proportion of products for which the order in which tasks begin differs from that of the preceding product of the same model. The second is the proportion of products for which worker–task assignments differ from those of the preceding product of the same model.

Complementarity of Managers' and Workers' Learning To test H3, we augment Equation 1 with a continuous interaction between the two experience measures:

$$Y_{jmt} = \alpha + \beta_1 \cdot \text{IHS}(\text{WorkerExp}_{w(jmt)}) + \beta_2 \cdot \text{IHS}(\text{ManagerExp}_{g(jmt)}) + \beta_3 \cdot \text{IHS}(\text{WorkerExp}_{w(jmt)}) \times \text{IHS}(\text{ManagerExp}_{g(jmt)}) + \gamma_m + \delta_t + \theta_w + \lambda_g + \varepsilon_{jmt} \quad (4)$$

The coefficient of interest, β_3, captures how the marginal return to worker experience changes as manager experience increases. Because lower values of these outcomes indicate better performance, a negative β_3 indicates complementarity: the performance improvement associated with one form of experience becomes larger as the other form of experience increases.

To test H4, we divide the product-level observations into sub-samples according to whether the manager combination associated with each product has high or low soft skills and estimate Equation 4 separately for each subsample. The soft-skill measure and classification criteria are described in Subsection 3.3.4.

Multiple Hypothesis Testing To account for the multiple statistical tests we conduct, we use the false discovery rate adjustment of Benjamini and Hochberg (1995) and report the resulting q-values in brackets below the standard errors. We apply this adjustment to the results for H1–H3, computing q-values jointly within families defined by hypothesis. For H1, the family contains eight coefficients: the worker- and manager-combination experience coefficients in each of the four outcome regressions. For H2, we form a separate family for each measure of managerial intervention. Each such family contains twelve coefficients: the two experience coefficients in each of the four outcome regressions, the interaction between worker experience and the intervention measure in the two execution-outcome regressions, and the interaction between manager experience and the intervention measure in the two coordination-outcome regressions. For H3, the family contains twelve coefficients: worker-combination experience, manager-combination experience, and their interaction in each of the four outcome regressions. For each individual test, the null hypothesis is that the corresponding coefficient equals zero, evaluated against a two-sided alternative.

4 RESULTS

4.1 Role Differentiation in Learning on Execution and Coordination

We first test the role-differentiated learning predicted by H1: worker shared experience should primarily improve execution performance, whereas manager shared experience should primarily improve coordination performance. Table 1 reports estimates based on Equation 1, with standard errors in parentheses and FDR-adjusted q-values in square brackets below. Worker experience primarily improves execution quality. Doubling worker-combination experience reduces the probability of a defect by 0.7 percentage points (q = 0.000), a 6.4% reduction relative to the mean defect rate of 10.8%, while its effects on operating time and

coordination outcomes are not statistically significant.[5] Manager experience, in contrast, primarily improves coordination performance: doubling manager-combination experience reduces waiting time by 2.7% (q = 0.054) and lowers the probability of material shortages by 0.59 percentage points (q = 0.028), a 1% reduction relative to the outcome mean, while it has no significant effect on either execution outcome.

INSERT TABLE 1 HERE

Benchmarking effect sizes We benchmark our estimated effect sizes against prior organizational learning studies. Figure A9 presents estimates from our paper alongside results from the literature (Levitt et al., 2013; Narayanan et al., 2009; Pisano et al., 2001; Reagans et al., 2005). Our role-specific estimates are generally smaller in magnitude than the aggregate learning effects documented in prior studies. However, aggregate estimates cannot identify how different organizational members contribute to the learning process. To illustrate this empirically, we separately regress each outcome on worker shared experience and manager shared experience, rather than including both simultaneously (Table A7). When the experience of each role is examined separately, worker experience appears to significantly reduce material shortages (Column 4), even though our joint specification attributes improvements in material shortages primarily to manager experience. Thus, examining only aggregate experience or the experience of a single organizational role can obscure the division of learning across roles.

Aggregate productivity validation To assess whether the four intermediate performance measures used to capture the specialization across organizational roles are consequential for overall productivity, we further examine how they relate to aggregate output.

[5]Experience enters the specification as IHS(X), and a doubling of X increases IHS(X) by approximately $\ln(2) \approx 0.693$. We therefore multiply each estimated coefficient by $\ln(2)$ to obtain the effect of a doubling of experience. For binary outcomes (defect, material shortage), this product is the resulting change in probability, reported in percentage points. For continuous outcomes that are themselves IHS-transformed (operating time, waiting time), this product approximates the percentage change in the outcome, since the IHS transformation behaves like the natural logarithm away from zero.

We regress daily output on the four performance measures, aggregating both to the model-day level. Appendix C.3 shows that higher daily output is associated with lower operating time, waiting time, and defect rates, but not with lower material shortage rates. Thus, both the reduction in defects from worker learning and the reduction in waiting time from manager learning translate into improvements in aggregate productivity.

4.2 Managerial Coordination Interventions and Learning

The role-differentiated learning results establish that worker and manager experience improve distinct dimensions of performance. We next test H2, which predicts that managers' active interventions in coordination activities strengthen both managers' coordination learning and workers' execution learning. To test this prediction, we extend the baseline specification in Equation 1 by interacting two measures of managerial intervention, task-reordering and worker-reallocation rates, with worker experience for the execution outcomes and with manager experience for the coordination outcomes (see Equation 2 and 3).

As shown in Column 1 of Table 2, there is no evidence that workers' learning in operating time is affected by managers' coordination interventions. In Column 2, workers show evidence of faster defect reduction under more active managerial interventions, although the effect loses statistical significance after the FDR adjustment. The non-interacted coefficient on worker experience remains negative and significant (q = 0.024), indicating that workers learn through repeated execution even when managerial reorganization is limited. Columns 3 and 4 show that in models with more frequent managerial interventions, manager experience is associated with faster improvement in coordination outcomes: the interaction with manager experience is negative and significant for both waiting time and material shortages (q = 0.001). The statistically insignificant non-interacted coefficients indicate that manager experience has no detectable association with coordination outcomes when the corresponding intervention rate is zero.

To address potential limitations in how these two measures capture managerial interven-

tion, we conduct two additional checks using alternative constructions. First, we combine task reordering and worker reallocation into composite measures using their average and minimum; Table A8 yields consistent findings. Second, our model-level intervention rates use changes observed across all products within a model and may confound adjustments with the pace of learning itself. We address this concern by reconstructing the measures using only changes observed in products completed prior to the focal product. Table A9 shows that the coordination result survives under this backward-looking reconstruction.

INSERT TABLE 2 HERE

4.3 Complementarity Across Learning Trajectories

Beyond role-differentiated learning, we next test H3, which suggests that experience in one role may increase the returns to experience in the other. On execution outcomes, we examine how manager experience shapes worker learning; on coordination outcomes, how worker experience shapes manager learning. We estimate complementarity between worker and manager experience based on Equation 4, with the estimates reported in Table A10. Figure 2 illustrates how the learning effect of each role varies with the experience accumulated in the other role.

Panel A of Figure 2 focuses on workers' learning on execution. Worker experience becomes increasingly effective at reducing operating time as manager experience accumulates, although the interaction does not remain statistically significant after the FDR adjustment ($q = 0.111$). However, workers' learning on defects remains statistically significant and almost unchanged across levels of manager experience. This contrast is consistent with H3, which predicts stronger complementarity for performance outcomes tied to production flow, where experienced managers can reduce coordination frictions that constrain workers' ability to apply what they know. In our setting, the firm's incentive structure may further reinforce this pattern, as workers are directly incentivized to maintain quality but have little direct incentive to improve speed, which depends on production processes beyond the individual

worker. Managers, responsible for production at a higher level, instead have a consistent incentive to improve speed and can translate this incentive into workers' execution.

For coordination outcomes, Panel B of Figure 2 shows that manager experience becomes increasingly effective at reducing waiting time as worker experience accumulates, whereas its effect on material shortages does not change significantly. Consistent with H3, this pattern suggests that for handoff-related coordination, more experienced workers make execution more predictable, allowing managers to translate their accumulated experience more effectively into reductions in waiting time.

INSERT FIGURE 2 HERE

4.4 How Managerial Soft Skills Enable Complementary Learning

Having established the complementarity between worker and manager experience, we next examine whether the cross-role interaction capacity strengthens the mutual reinforcement of learning. As described in Section 3.3.4, we construct PCA- and factor-analysis-based indices of managerial soft skills at the model level and classify models as having high or low managerial soft skills based on a median split of the sample. We estimate the complementarity based on Equation 4 separately in these two subsamples and report results in Table 3.[6] For execution, H4 leaves open whether stronger managerial soft skills improve workers' learning conditions or make workers rely more on managers instead of developing execution routines independently. The evidence is more consistent with the latter. For operating time, the complementarity effect is smaller in magnitude in the high-soft-skills subsample (−0.007) than in the low-soft-skills subsample (−0.014), while the effects on defects are nearly identical across groups. For coordination, by contrast, the complementarity effects are larger in magnitude in the high-soft-skills subsample than in the low-soft-skills subsample. For waiting time, the estimates are −0.031 versus −0.011, and for material shortages, −0.006 versus −0.003.

[6] We focus on the PCA-based results in Table 3 and also report the factor-analysis-based results in Table A11 as a robustness check.

These directional differences are consistent with H4, but should be interpreted cautiously. Formal z-tests in Table 3 do not reject equality of the interaction coefficients across high- and low-soft-skills subsamples for any outcome. We therefore treat the results as exploratory rather than conclusive evidence.

INSERT TABLE 3 HERE

4.5 Robustness Checks

We conduct several additional analyses to assess the robustness of our findings on role-differentiated learning (H1), the facilitating role of managerial intervention (H2), and complementarity in learning trajectories (H3). First, the estimated patterns are not driven by experience accumulated outside the focal team or model. We separately control for members' individual model-specific experience (Figures A11, A13, and A15) and the team combination's cumulative production volume across all models (Figures A12, A14, and A16), including each control continuously or through percentile bins that partition its distribution into three, five, and ten groups. The results remain broadly robust, although under finer controls for individual model-specific experience, the manager-experience effect on waiting time and the complementarity between worker and manager experience in waiting time become less statistically stable. Second, our findings are not driven by selective team dissolution, whereby poorly functioning teams may be more likely to dissolve, jointly affecting experience accumulation and performance outcomes. The results remain robust when controlling for lagged worker turnover under each manager combination (Tables A12–A15). Third, our results are not sensitive to the choice of functional form. Replacing the IHS transformations of operating time, waiting time, and experience with logarithmic transformations yields qualitatively similar results (Tables A16–A18).

5 DISCUSSION AND CONCLUSION

This study shows what aggregate measures of team experience conceal: organizational learning in hierarchical production teams is role differentiated. Workers and middle managers do not accumulate interchangeable experience. Worker shared experience is associated with fewer defects, while manager shared experience is consistently associated with fewer material shortages and generally associated with shorter waiting time. The waiting time relationship, however, is sensitive to flexible controls for members' other model-specific experience. Managerial intervention can also facilitate managers' learning in coordination. We also find cross-role complementarity, particularly for production flow related outcomes, where performance depends on linkages across roles: worker experience strengthens managers' learning in reducing waiting time, an interaction that survives the Benjamini–Hochberg adjustment ($q = 0.064$), while manager experience strengthens workers' learning in reducing operating time, although this interaction does not survive the adjustment ($q = 0.111$). The soft-skills results are less conclusive: although the coefficients point toward stronger coordination complementarities when managerial soft skills are high, formal tests do not reject equality across the high- and low-skill subsamples. We therefore treat H4 as exploratory. Together, these findings show that organizational learning depends on which role accumulates experience, which performance problem it addresses, and how experience is combined across roles.

The findings advance the study of organizational learning in four ways. First, we reveal how learning is distributed across roles within teams. Prior work often measures experience at the team, plant, or firm level. Such measures combine the experience of members who perform different work and receive different feedback. Our data identify the worker and manager combinations assigned to the same product, allowing us to distinguish learning associated with execution from learning associated with coordination.

Second, our study shows why it is necessary to document learning effects across multiple performance dimensions within the same setting. A single aggregate learning curve would obscure important underlying mechanisms, such as reducing execution errors, accelerating

task completion, smoothing handoffs, or improving access to materials. It would also make findings across studies difficult to interpret when their dependent variables capture different processes (Harvey et al., 2022). By examining execution, coordination, and aggregate output within the same production system, we show that production experience maps onto different dimensions of performance depending on who accumulates it.

Third, the findings show that differentiated learning can be integrated across different roles within teams or organizations. Worker and manager experience are complementary when performance depends on both stable task execution and effective production flow. The evidence is strongest for waiting time and more tentative for operating time, so the value of complementarity is outcome specific rather than universal. Going beyond prior studies which focus mainly on the effects of managers on workers, we show the understudied reverse direction: workers' experience can in turn facilitates manager's learning.

Finally, we unpack one channel through which middle managers shape learning across roles. We find that managers who more actively redesign task sequences and reallocate work facilitate faster learning among both themselves and their subordinates. The same manager combinations intervene at similar frequencies across models, indicating a persistent managerial style rather than model-specific demands and reinforcing our interpretation that coordination interventions accelerate learning. This result complements evidence that managerial characteristics affect worker learning by distinguishing managers' accumulated experience from the conditions under which that experience is applied (Adhvaryu et al., 2023). To more clearly articulate our contribution to and position within the organizational learning literature, Table A19 and Figure A17 in Appendix H provide a detailed comparison of prior research.

The findings suggest several ways firms can manage learning across roles. Firms should first diagnose the performance bottleneck before deciding which type of experience to preserve. When defects are the primary concern, retaining workers' model-specific shared experience may be valuable. When handoffs or production flow constrain performance, retaining

relevant managerial coordination experience may matter more. Firms can also use digital production and human-resource systems to track the shared experience accumulated by particular worker and manager combinations with particular products. Such records can inform assignment, retention, and rotation decisions more directly than individual tenure or aggregate team output alone. Finally, firms should treat experience as a relational resource. Because experience in one role can increase the returns to experience in another, firms may act sub-optimally when team-formation and retention decisions fail to consider how experience is distributed and combined across interdependent roles.

Our study has several limitations. First, our empirical analysis examines a single manufacturer in which responsibilities are allocated between frontline workers and team managers in the way described above. This allocation may differ in other settings, producing different empirical patterns of role-differentiated learning. For example, when workers have greater autonomy, their experience may also improve coordination performance. Managerial experience should matter more when production requires frequent coordination and managers can adjust assignments and workflows, but less when tasks are independent or coordination is prescribed by formal systems. However, this does not undermine the validity of our framework, whose central prediction is that differences in responsibilities, actions, and feedback across roles shape the division of learning, not that particular roles always improve particular dimensions of performance. Besides, although shared experience can improve role-specific routines and make behavior more predictable across hierarchical levels, the same repetition that supports learning can also impose costs. It may entrench inefficient routines, narrow exposure to alternative practices, and reduce flexibility when production conditions change. Future research could examine when the learning gains from preserving shared experience exceed the benefits of knowledge diffusion across teams and the operational flexibility that timely reassignment can provide.

AI Use Statement. The authors used ChatGPT (OpenAI) and Claude (Anthropic) to assist with language editing, including improving the grammar and clarity of written text,

and with checking programming code. All AI-assisted output was reviewed, verified, and edited by the authors. The authors take full responsibility for the accuracy and integrity of all material in the manuscript.

Figures and Tables

Figures

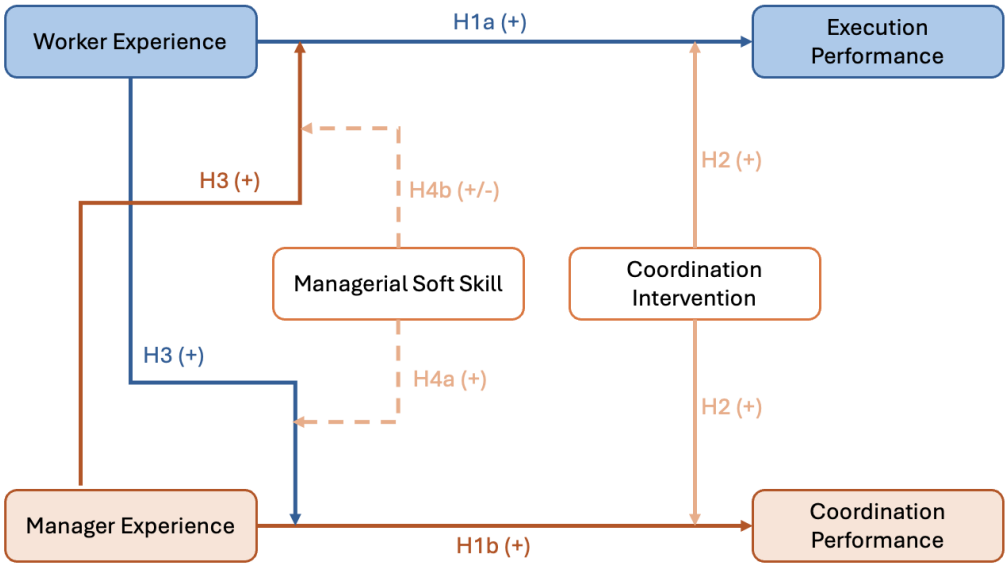
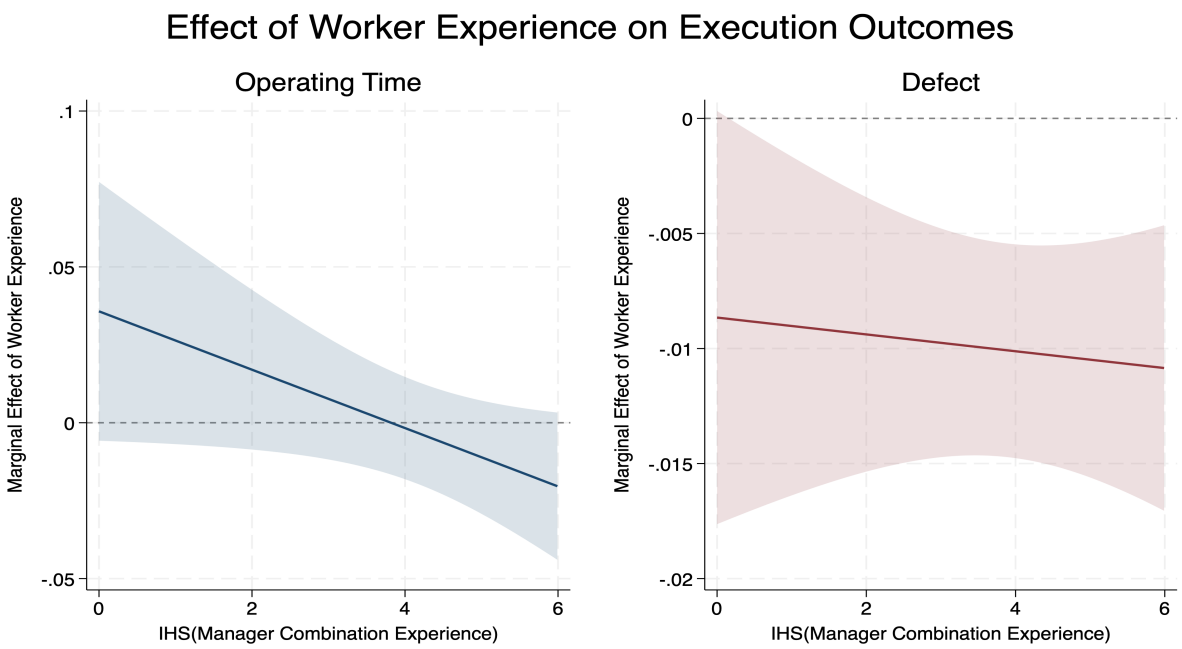
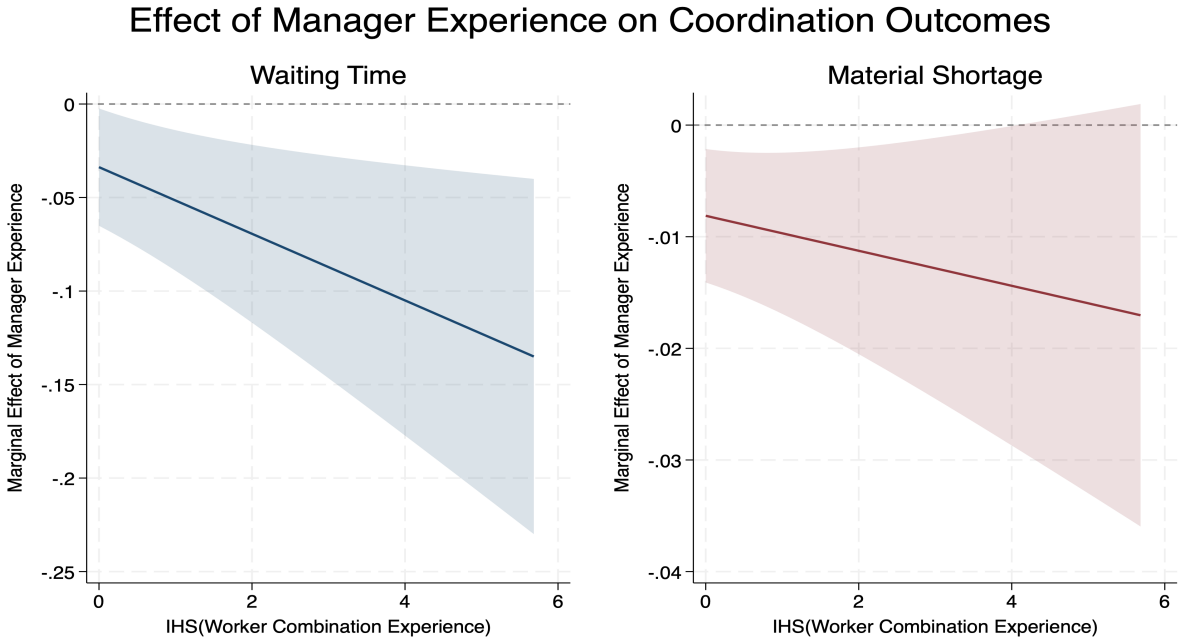


Figure 1: Theoretical framework and hypotheses (H1–H4).

Notes: Figure 1 presents our theoretical framework. The left-hand side displays the two experience variables, worker experience and manager experience, while the right-hand side displays the two dimensions of performance, execution and coordination. Solid lines denote predictions that receive clear empirical support, whereas dashed lines denote predictions for which the evidence is less robust.



(a) How manager experience affects workers' learning in execution



(b) How worker experience affects managers' learning in coordination

Figure 2: Complementarity between Worker and Manager Experience

Notes: Panel (a) plots how the marginal effect of worker experience on execution outcomes changes with manager experience; Panel (b) plots how the marginal effect of manager experience on coordination outcomes changes with worker experience. Shaded areas denote 95% confidence intervals.

Tables

Table 1: Effects of Worker and Manager Experience on Performance

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
IHS(Exp. of Wkr. Combn.)	0.000	-0.010	0.014	0.003
	(0.009)	(0.002)	(0.013)	(0.003)
	[0.971]	[0.000]	[0.364]	[0.393]
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	-0.039	-0.009
	(0.011)	(0.003)	(0.017)	(0.003)
	[0.364]	[0.364]	[0.054]	[0.028]
P-value (H0: effect of worker = effect of manager)	0.462	0.004	0.053	0.041
Dependent Variable Mean	1.221	0.108	2.556	0.654
Observations	76,690	76,690	76,690	76,690
R-squared	0.761	0.343	0.614	0.570
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: The dependent variables include the inverse hyperbolic sine (IHS) of total operating time, a binary indicator for whether a production defect occurred, the IHS of waiting time between tasks, and a binary indicator for whether a material shortage occurred. The key independent variables are the IHS of cumulative production experience for both the worker combination and the manager combination involved in producing the product, measured by the number of completed units prior to the current one. All regressions include fixed effects for product model, calendar month, and the specific worker and manager combinations involved in production. Standard errors, clustered at the product model level, are reported in parentheses. Benjamini–Hochberg FDR-adjusted q-values, computed jointly across the eight main experience coefficients in this table (worker- and manager-combination experience across the four outcomes), are reported in brackets.

Table 2: Managerial Coordination Interventions and Learning Effects

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time) (1)	Defect (2)	IHS(Waiting Time) (3)	Material Shortage (4)
Panel A: Task Reordering				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.007	0.013	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
	[0.595]	[0.024]	[0.435]	[0.458]
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	0.001	-0.001
	(0.011)	(0.003)	(0.017)	(0.003)
	[0.396]	[0.396]	[0.966]	[0.845]
IHS(Exp. of Wkr. Combn.)× Rate of Task Reordering	0.055	-0.023		
	(0.038)	(0.013)		
	[0.363]	[0.243]		
IHS(Exp. of Mgr. Combn.)× Rate of Task Reordering			-0.299	-0.059
			(0.074)	(0.015)
			[0.001]	[0.001]
Observations	75,762	75,762	75,762	75,762
R-squared	0.752	0.332	0.607	0.567
Panel B: Worker Reallocation				
IHS(Exp. of Wkr. Combn.)	-0.006	-0.007	0.013	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
	[0.639]	[0.030]	[0.442]	[0.465]
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	0.002	-0.000
	(0.011)	(0.003)	(0.017)	(0.003)
	[0.398]	[0.398]	[0.892]	[0.892]
IHS(Exp. of Wkr. Combn.)× Rate of Worker Reallocation	0.050	-0.025		
	(0.039)	(0.013)		
	[0.398]	[0.199]		
IHS(Exp. of Mgr. Combn.)× Rate of Worker Reallocation			-0.308	-0.062
			(0.075)	(0.015)
			[0.000]	[0.000]
Observations	75,762	75,762	75,762	75,762
R-squared	0.752	0.332	0.607	0.567
Dependent Variable Mean	1.221	0.108	2.556	0.654
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: The dependent variables include the IHS of total operating time, a binary indicator for defect, the IHS of waiting time, and a binary indicator for material shortage. Panel A interacts worker and manager experience with the Rate of Task Reordering, defined as the share of products within a model for which the task sequence changes relative to the prior product. Panel B interacts with the Rate of Worker Reallocation, defined as the share of products for which worker-to-task assignments change. All four measures are model-level. All regressions include model, year-month, worker combination, and manager combination fixed effects. Standard errors, clustered at the product model level, are reported in parentheses. Benjamini–Hochberg FDR-adjusted q-values, computed jointly across the twelve experience-related coefficients within each panel (worker experience, manager experience, and the interaction across the four outcomes), are reported in brackets.

Table 3: Heterogeneity of Complementarity Effects: Managerial Soft Skills (PCA)

Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Operating Time)		Defect		IHS(Waiting Time)		Material Shortage	
	Managerial Soft Skills		Managerial Soft Skills		Managerial Soft Skills		Managerial Soft Skills	
	High	Low	High	Low	High	Low	High	Low
IHS(Exp. of Wkr. Combn.)	0.037	0.043	-0.004	-0.003	0.191	0.035	0.043	0.012
	(0.046)	(0.026)	(0.010)	(0.007)	(0.074)	(0.038)	(0.014)	(0.007)
IHS(Exp. of Mgr. Combn.)	-0.064	0.024	-0.008	0.005	-0.078	-0.034	-0.028	-0.008
	(0.027)	(0.012)	(0.006)	(0.004)	(0.042)	(0.020)	(0.008)	(0.004)
IHS(Exp. of Wkr. Combn.) * IHS(Exp. of Mgr. Combn.)	-0.007	-0.014	-0.002	-0.002	-0.031	-0.011	-0.006	-0.003
	(0.010)	(0.005)	(0.003)	(0.001)	(0.017)	(0.008)	(0.003)	(0.002)
P-value for equality of Interaction effects (High vs. Low subsamples)	0.522		0.868		0.297		0.475	
Dependent Variable Mean	1.970	1.048	0.145	0.104	2.933	2.558	0.561	0.650
Observations	15,654	45,312	15,654	45,312	15,654	45,312	15,654	45,312
R-squared	0.865	0.759	0.499	0.335	0.740	0.640	0.629	0.620
Year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables include the inverse hyperbolic sine (IHS) of total operating time, a binary indicator for whether a production defect occurred, the IHS of waiting time between tasks, and a binary indicator for whether a material shortage occurred. Columns (1)–(8) split the sample by a PCA-based index of managerial soft skills constructed from workers' survey evaluations. The index summarizes five dimensions: responsiveness, problem diagnosis, problem-solving, understanding of workers, and communication. Standard errors are clustered at the product-model level.

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Online Appendix

A Data and Variable Construction

A.1 Dependent Variables in Prior Literature

Table A1: Dependent Variables Used in This Study and Related Literature

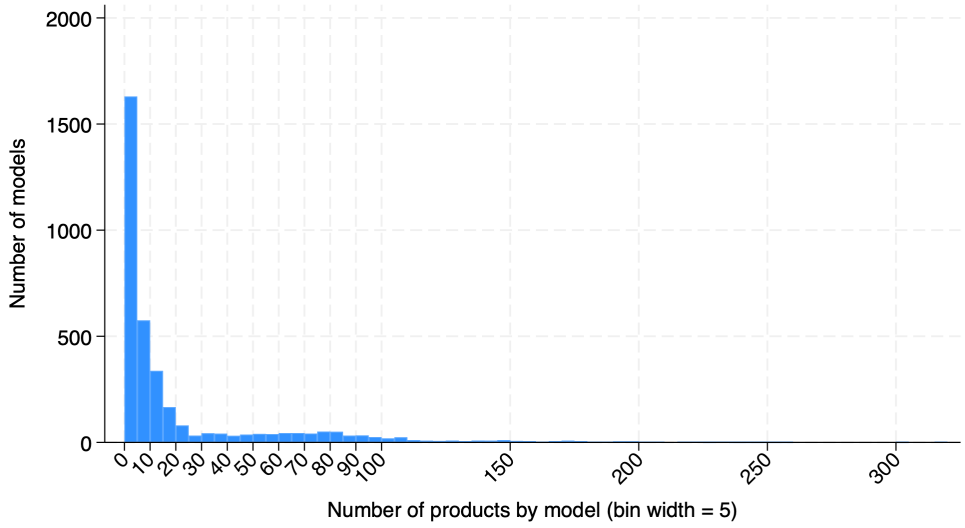
Our DV	Similar DV	Paper
Operating time	duration per unit	Thurstone (1919) Reagans et al. (2005) Espinosa et al. (2007) Narayanan et al. (2009) Ramdas et al. (2018)
Defects	30-day mortality of patients	Chen (2021)
	defects per product	Levitt et al. (2013)
	returns and ratings	Hayward (2002)
	RARW	Clark et al. (2018)
	abnormal returns	Anand & Khanna (2000)
	defect density	Hatch & Mowery (1998) Hatch & Dyer (2004)
	deviation of radiologists	Ibanez et al. (2018)
	product quality	Eggers (2012)
Productivity	labor requirements (unit cost)	Wright (1936) Alchian (1963) Thornton & Thompson (2001) Jain (2013)
	unit cost	Epple et al. (1996)
	labor hours per unit (unit cost)	Benkard (2000) Fong Boh et al. (2007)
	output per period	Schilling et al. (2003) Balasubramanian & Lieberman (2010) Kellogg (2011) Levitt et al. (2013) Egelman et al. (2017) Adhvaryu et al. (2023) Ilzetzki (2024) Chan et al. (2014) Haggag et al. (2017)

Notes: This table summarizes the literature in organizational learning using similar dependent variables as those in our analysis. For operating time, we include (individual) output per period because that's just the reciprocal of task operating time. For defects, we include all kinds of measures for production quality because in different context it can be very different. For productivity, we include all kinds of measure showing the overall production efficiency. To our knowledge, there's no paper in organizational learning using measures similar to waiting time or material shortage.

A.2 Structure of Production Data

To illustrate the three-level (model–product–task) data structure in our context, consider a specific power transmission unit produced by this firm, which we refer to as Model X. During our observation window, Model X was produced five times, yielding five individual products. Producing each product required a sequence of tasks: frame welding, component installation, circuit wiring, quality inspection, and final delivery preparation. These tasks were carried out by workers under the supervision of their respective managers. For each task, our digital tracking system records the start time, end time, and the assigned worker. These task-level records are then aggregated to the product level to construct our outcome variables. Figure A1 shows the distribution of products across all models in our sample.

Figure A1: Distribution of the Number of Products by Model



Notes: This figure shows the distribution of the number of products produced for each model during the sample period. Each bar represents a range of product counts (bin width = 5). The distribution is highly skewed, with many models produced only a few times and a small number of models produced repeatedly.

A.3 Survey Coverage and Attrition

The managerial soft-skills survey was administered in September 2024 and covers 156 of the 299 workers who appear in the production data between January 2022 and June 2024; the remaining 143 had left the firm by the survey date. To assess whether this gap reflects selective attrition, we first compare respondents and non-respondents on observable baseline characteristics (gender, education, work experience, ethnicity, and wages) and find no statistically significant differences. We then test whether attrition is related to unobserved team-level conditions, such as team conflict that could simultaneously worsen production outcomes and induce workers to leave. Using a manager-combination–by-month panel, we regress each combination's worker exit rate in the following month on its current-month production outcomes. As shown in Table A2, none of the four outcomes predicts subsequent worker exit, suggesting that survey attrition is not systematically related to either observable worker characteristics or team performance.

Table A2: Team Production Outcomes and Subsequent Worker Attrition

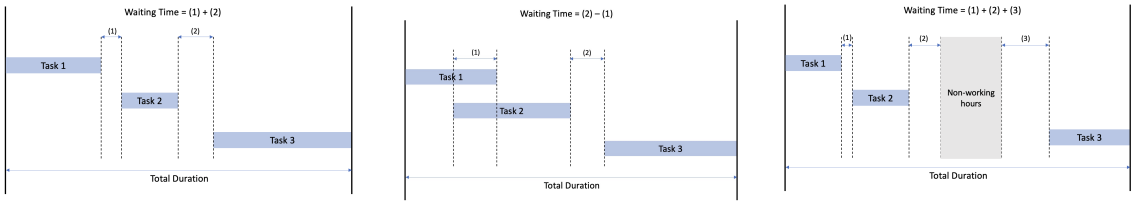
	(1)	(2)	(3)	(4)	(5)
VARIABLES	Worker Exit Rate (t+1)	Worker Exit Rate (t+1)	Worker Exit Rate (t+1)	Worker Exit Rate (t+1)	Worker Exit Rate (t+1)
IHS(Operating Time)	-0.001				-0.001
	(0.001)				(0.001)
Defect		-0.000			-0.000
		(0.003)			(0.003)
IHS(Waiting Time)			0.000		-0.000
			(0.000)		(0.000)
Material Shortage				0.002	0.002
				(0.002)	(0.002)
Observations	1,210	1,210	1,210	1,210	1,210
R-squared	0.202	0.200	0.200	0.201	0.202
Yearmonth FE	Yes	Yes	Yes	Yes	Yes
Cluster	Manager Combination	Manager Combination	Manager Combination	Manager Combination	Manager Combination
Dependent Variable Mean	0.0167	0.0167	0.0167	0.0167	0.0167

Notes: This table tests whether manager combinations experiencing worse production outcomes are more likely to lose workers subsequently. Each observation is a manager combination–month pair. The dependent variable is the combination's worker exit rate in the following month, defined as the share of workers employed under the combination's member group leaders in month $t + 1$ whose last appearance in the personnel records occurs in that month; workers appearing under multiple member leaders of the same combination are counted once, and the final sample month is excluded from the exit definition. Explanatory variables are monthly means across all products completed by the combination in month t. Columns (1)–(4) enter each outcome separately; Column (5) includes all four jointly. All specifications control for the monthly mean of IHS(workload) and include year-month fixed effects. Standard errors, clustered by manager combination, are in parentheses.

A.4 Definition of Waiting Time

Figure A2: Definition of Waiting Time

(a) No Task Overlapping (b) With Task Overlapping (c) Cross-Day



Notes: This figure illustrates the calculation of inter-task waiting time under three scenarios: (a) no task overlapping, where tasks are performed sequentially; (b) task overlapping, where tasks partially overlap in execution; and (c) cross-day production, where tasks span multiple working days with non-working hours subtracted. In all cases, waiting time is defined as the difference between the total production duration and the sum of individual task durations.

A.5 Construction of Experience Variables

We define experience separately for worker and manager combinations. This distinction allows us to isolate role-specific experience and examine whether worker and manager learning contribute differently across outcomes. To illustrate the construction of the experience variables, consider a hypothetical example in which Managers A and B are jointly assigned to a product, while Workers 1 and 2 are initially assigned to this product. Figure A3 provides a graphical illustration of this hypothetical example across ten successive products of the same model. For Products 4 through 6, the managers selects Worker 3 instead of Worker 2, creating a new Worker Combination {1,3}. The experience of this new worker combination begins at zero, while Manager Combination {A,B} continues to accumulate experience without interruption. From Product 7 onward, the original Worker Combination {1,2} returns and resumes from its previously accumulated experience level rather than restarting from zero. The same logic applies when the membership of a manager combination changes. Figure A4 then shows how this construction appears in the actual production data, using two representative models.

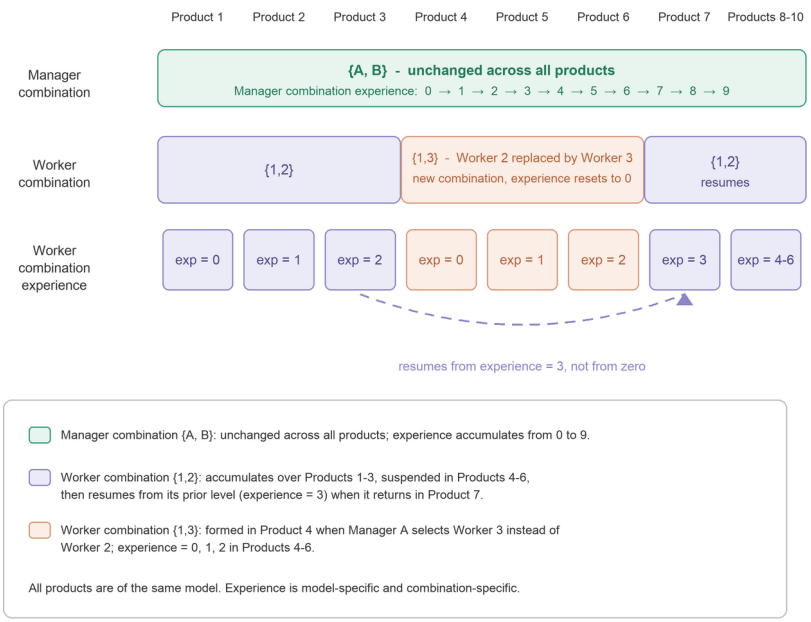
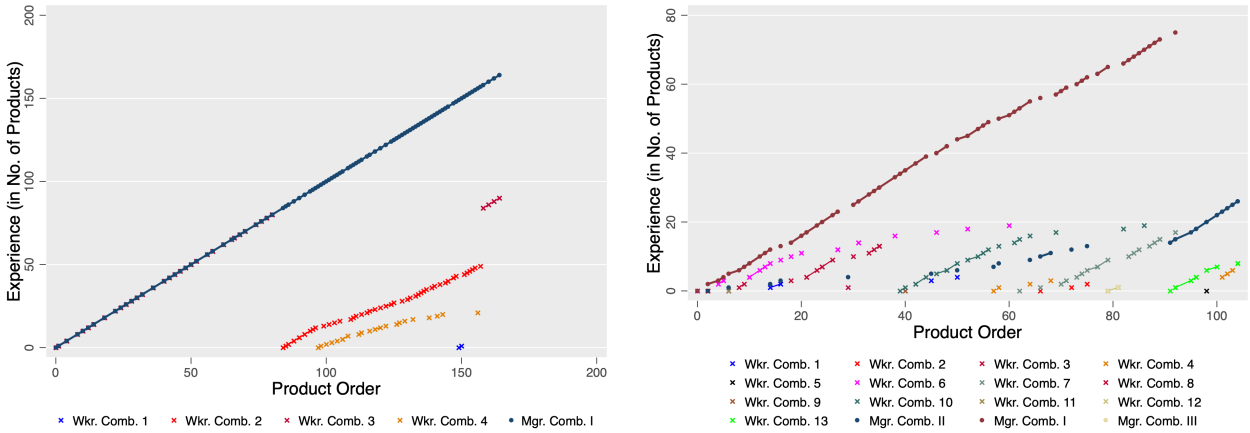


Figure A3: Worker and Manager Experience: An Example Illustration



(a) (b)

Figure A4: Worker and Manager Experience in the Production Data

Notes: This figure illustrates how ourconstruction of experience appears in the actual production data. Experience is measured as the cumulative number of same-model products previously completed by a given worker or manager combination. Experience resets to zero only the first time a given combination appears in the production sequence for that model. If a combination that previously worked together is replaced and later reappears, its experience resumes from the level it had previously accumulated rather than restarting from zero. Panel (a) shows a case where the manager combination remains fixed, while worker combinations vary, leading to different worker experience accumulation paths. Panel (b) depicts a scenario where both worker and manager combinations change, resulting in more frequent resets and greater dispersion in experience accumulation.

A.6 Construction of Task-Reordering and Worker-Reallocation Rates

We define task reordering based on whether the sequence of production tasks changes across consecutive products within the same model. For example, if the first product follows the order "Task A, Task B, and Task C," while the second product follows "Task A, Task C, and Task B," we code the second product as having task reordering. For example, if a model contains 10 products and 2 of them involve task reordering, the task-reordering rate equals 20%. We define worker reallocation based on whether the assignment of workers to tasks changes across consecutive products within the same model. For example, if the first product assigns Worker A to Task A and Worker B to Task B, while the second product assigns Worker B to Task A and Worker A to Task B, we code the second product as having worker reallocation.

Task reordering and worker reallocation capture two conceptually distinct managerial decisions: changing the order in which tasks are performed, and changing which worker is assigned to which task. Empirically, the two rates are strongly correlated, because one common managerial adjustment changes both at once: reorganizing an interleaved task sequence, in which workers alternate across tasks, into a blocked sequence, in which each worker handles a continuous set of tasks. The two measures are not equivalent, however, since each also moves independently when managers change only the task order or only the worker assignment. Figure A5 plots this relationship.

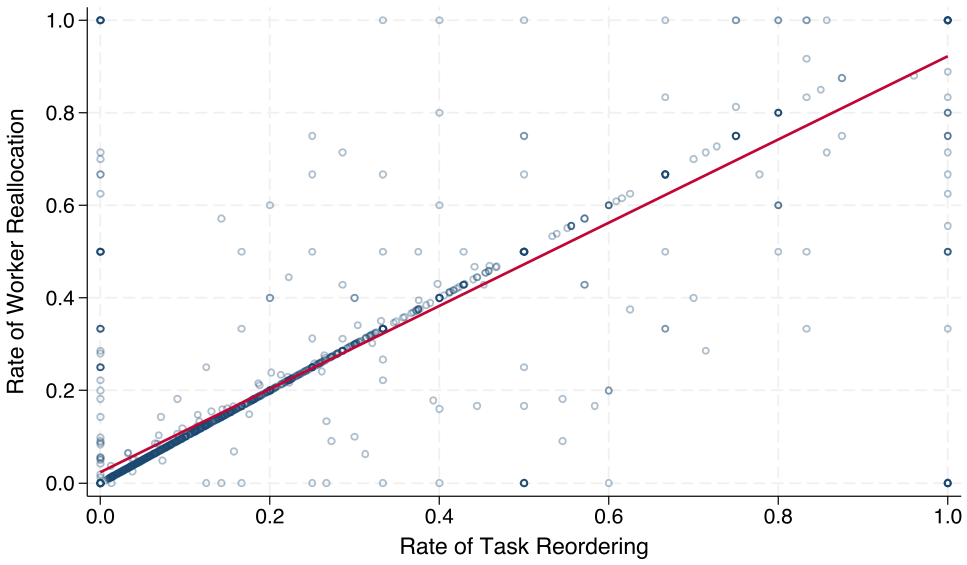


Figure A5: Correlation between Task-Reordering and Worker-Reallocation Rates

Notes: Each point is one product model. The horizontal axis is the model's task-reordering rate and the vertical axis its worker-reallocation rate, each defined as the share of products within the model that exhibit the corresponding change. The solid line is the fitted regression line.

A.7 Construction of Managerial Soft Skills

We construct a measure of managerial soft skills using survey responses from workers. The index is based on five items capturing different dimensions of managerial behavior: understanding of workers, quality of manager–worker communication, accuracy of problem diagnosis, speed of feedback, and problem-solving ability. The index reflects the interaction quality between managers and workers regarding communication and problem-solving.

To summarize these dimensions into a single index, we apply principal component analysis (PCA). Comparing the PCA and factor-analysis results, the first component/factor is clearly dominant under both approaches, with the first principal component in the PCA explaining over 80% of the total variation and subsequent components contributing little additional explanatory power (Figure A6). The loading comparison likewise shows that all survey items load positively and strongly under both methods, with highly similar patterns across PCA and factor analysis (Figure A7). The two indices are also highly correlated at the observation level, as shown in Figure A8. We therefore use the PCA-based composite index

as our primary measure of managerial soft skills in the main analysis and employ the factor-analysis-based index as a robustness check.

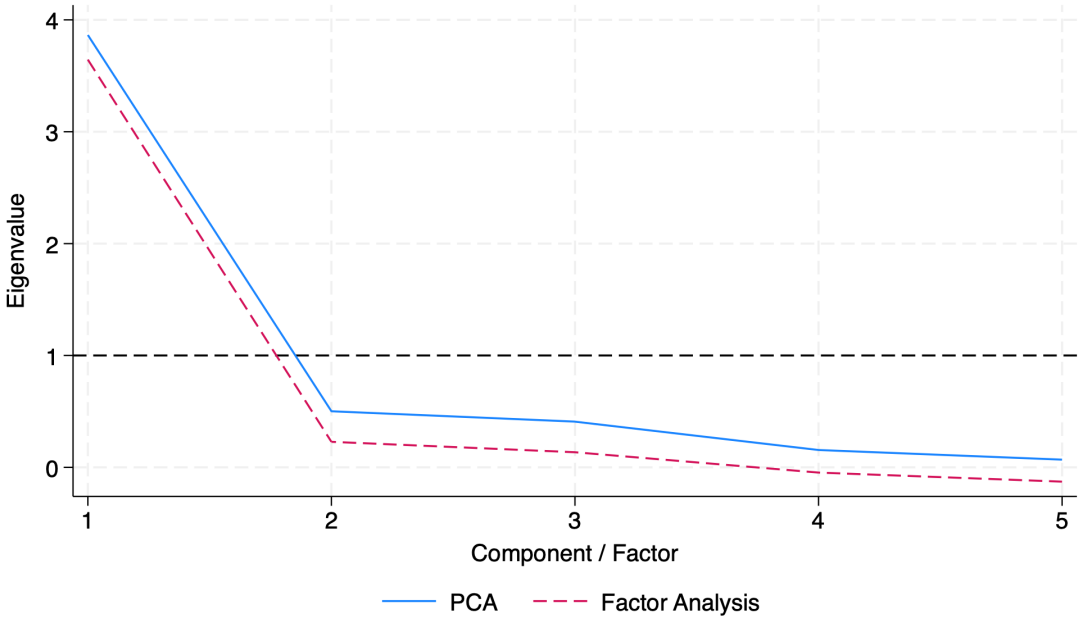


Figure A6: Comparison of PCA and Factor Analysis: Importance of the First Component

Notes: This figure compares how much of the overall variation in the five survey items is captured by each component under PCA and factor analysis. In both approaches, the first component/factor is much more important than the others. In the PCA, the first component alone explains more than 80% of the total variation across the five items, while each additional component contributes only a small amount. This pattern suggests that the five survey items largely reflect a single underlying dimension of managerial soft skills.

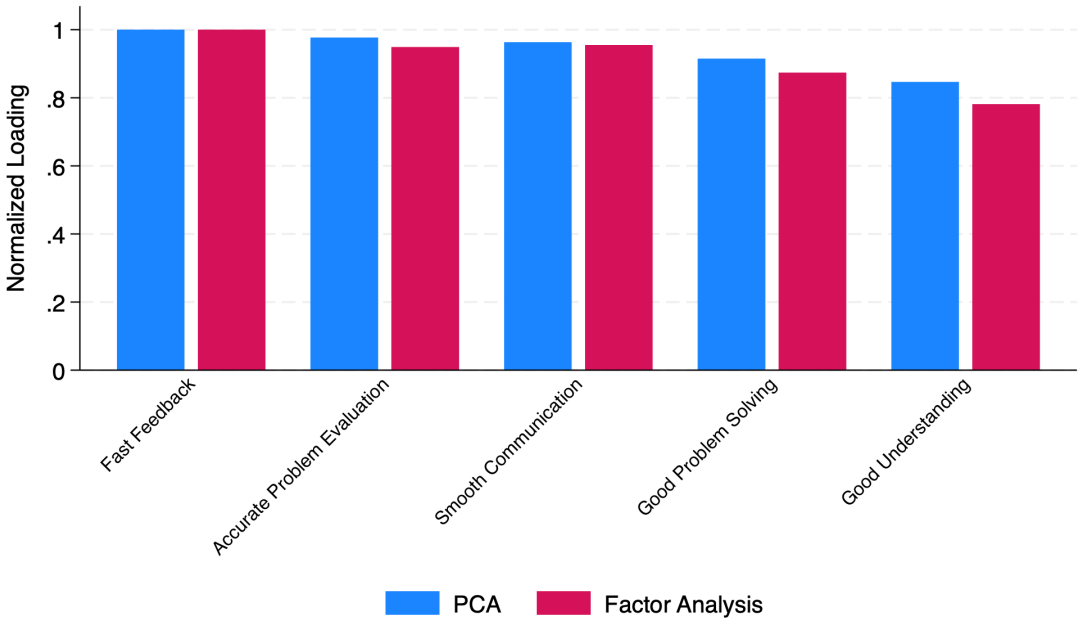


Figure A7: Comparison of PCA and Factor Analysis: Contribution of Each Survey Item

Notes: This figure compares the normalized loadings of each survey item under principal component analysis (PCA) and factor analysis, where the loading reflects the contribution of each survey item to the common measure of managerial soft skills represented by the first component or factor. Loadings are normalized such that the maximum loading within each method equals one, which facilitates comparison of relative patterns across items, as the raw magnitudes are not directly comparable across PCA and factor analysis due to different scaling conventions. Both methods yield highly similar loading patterns across the five survey items.

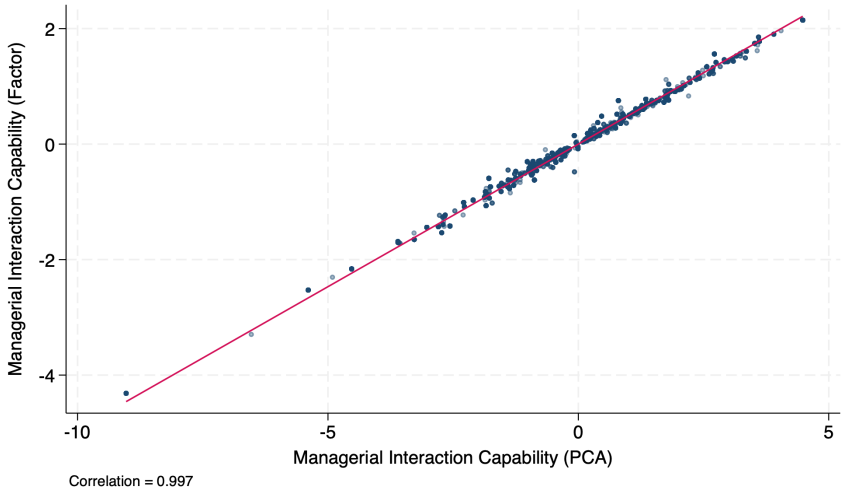


Figure A8: Correlation Between PCA and Factor Analysis Scores

Notes: This figure plots the observation-level Managerial Soft Skills scores derived from PCA (horizontal axis) against those derived from factor analysis (vertical axis). Each point represents one product observation. The high correlation between the two scores confirms that the PCA and factor-analysis approaches capture the same underlying construct.

A.8 Summary Statistics

Table A3: Descriptive Statistics

Variable	Mean	SD	p10	p25	p50	p75	p90
Panel A. Model Level (N = 3,556)							
Number of Products per Model	22.396	38.431	2	2	6	18	77
Number of Worker-Combination Changes	7.562	13.325	0	0	2	9	23
Number of Manager-Combination Changes	4.576	8.635	0	0	1	5	15
Managerial Interventions							
Rate of Task Reordering	0.211	0.334	0	0	0	0.286	1
Rate of Worker Reallocation	0.212	0.332	0	0	0	0.286	1
Panel B. Product Level (N = 79,639)							
Dependent Variables							
Operating Time (Hours)	6.840	27.801	0.017	0.067	0.650	3.998	15.567
Whether Defect Occurred	0.108	0.310	0	0	0	0	1
Waiting Time (Hours)	26.264	64.113	-0.004	1.423	9.890	25.279	68.823
Whether Material Shortage Occurred	0.654	0.476	0	0	1	1	1
Independent Variables							
Experience of Worker Combination (Product Counts)	5.948	11.865	0	0	1	7	17
Experience of Manager Combination (Product Counts)	16.396	25.465	0	0	5	23	47

Notes: This table reports summary statistics of the main variables used in the analysis. Panel A reports model-level variables. A worker-combination (manager-combination) change occurs when the worker (manager) combination of a product differs from that of the preceding product of the same model in production order. The rates of task reordering and worker reallocation compare consecutive products made by the same worker combination within a model; they are therefore defined only for the 2,765 of the 3,556 models in which at least one worker combination produced more than one product, and these models account for 97.3% of all products. Panel B reports product-level variables: the dependent variables (operating time, defects, waiting time, and material shortages) and the key independent variables, which capture the cumulative production experience of worker and manager combinations. The first two columns present respectively the mean and standard deviations. The last five columns report the values of the 10th, 25th, 50th, 75th, and 90th percentiles.

B Identification Strategy

Tables A4, A5, and A6 together support the identification assumption that variation in experience accumulation is not driven by selection on past performance.

Tables A4 and A5 test whether lagged performance predicts future production assignments at the team and worker level, respectively. In both tables, the dependent variables are the number of new models allocated, average output volume per model, and average workload per model in a given month. The regressors include lagged workload, lagged output or wages, and worker or team characteristics. Coefficients are generally small and statistically insignificant, and joint F-statistics and low R^2 values indicate that past performance has little predictive power over future assignments.

Table A6 addresses a related concern: whether teams with stronger early performance are systematically retained for longer production sequences, which would confound experience accumulation with selection. The dependent variable is the total number of distinct models produced by each team over the sample period. Early performance is measured using average output and defect rates across a team's first two (column 1) or first three (column 2) completed models. Neither measure significantly predicts total production involvement, supporting the view that retention is not driven by early performance.

Table A4: Predictability of Future Production Assignments Based on Current Production Characteristics

Dependent Variables	Number of New Models (1)	Production Volumes per Model (2)	Workload per Model (3)
Lagged Mean of Workload per Model	0.011	0.005	0.008
	(0.025)	(0.004)	(0.036)
Lagged Team Output	-0.004	0.000	0.006
	(0.003)	(0.000)	(0.004)
Lagged Manager Bonus	0.012	0.001	-0.009
	(0.021)	(0.003)	(0.031)
Team Size	1.487	0.012	1.038
	(0.450)	(0.073)	(0.667)
Worker %Female	-40.619	1.013	11.350
	(36.360)	(5.884)	(53.878)
Worker %Ethnic Han	-25.406	-1.415	-16.969
	(26.464)	(4.283)	(39.214)
Worker % College	4.128	-2.535	-38.561
	(28.394)	(4.595)	(42.074)
Year-month FE	Yes	Yes	Yes
Team FE	Yes	Yes	Yes
Observations	272	272	272
Adjusted Within R-square	0.049	-0.017	-0.003

Notes: This table presents regression results examining whether current production characteristics can predict future production assignments at the team level. The dependent variables include key indicators of production assignments: the number of new models allocated (column 1), the average output volume per model (column 2), and the average workload per model (column 3). Independent variables include production characteristics from the previous month, such as the lagged mean workload per model, lagged team output (proxied by the average monthly wages of workers), and lagged manager bonuses. In addition, we control for team characteristics, including team size (i.e., the number of workers), the proportion of female workers, the proportion of Han ethnicity workers, and the proportion of workers with a college degree. All regressions control for team fixed effects and year-month fixed effects.

Table A5: Predictability of Future Production Assignments Based on Worker's Current Production Characteristics

Dependent Variables	Number of New Models (1)	Production Volumes per Model (2)	Workload per Model (3)
Lagged Mean of Workload per Model	-0.003	0.007	-0.030
	(0.008)	(0.005)	(0.042)
Lagged Monthly Wage	-0.001	0.001	-0.002
	(0.001)	(0.001)	(0.004)
Worker Experience	-0.428	0.453	-4.263
	(2.384)	(1.617)	(12.964)
Year-month FE	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes
Observations	1,139	1,139	1,139
Adjusted Within R-square	-0.002	0.001	-0.002

Notes: This table presents regression results testing whether a worker's current production characteristics predict future production assignments. The dependent variables capture key aspects of assignment: the number of new models allocated to the worker (column 1), the average output volume per model (column 2), and the average workload per model (column 3) in a given month. The main independent variables are lagged production characteristics from the previous month, including the worker's mean workload per model and monthly wage. We also control for worker experience, measured by the number of years the worker has been employed at the firm. All regressions include worker fixed effects and year–month fixed effects.

Table A6: Early Team Performance and Subsequent Production Engagement

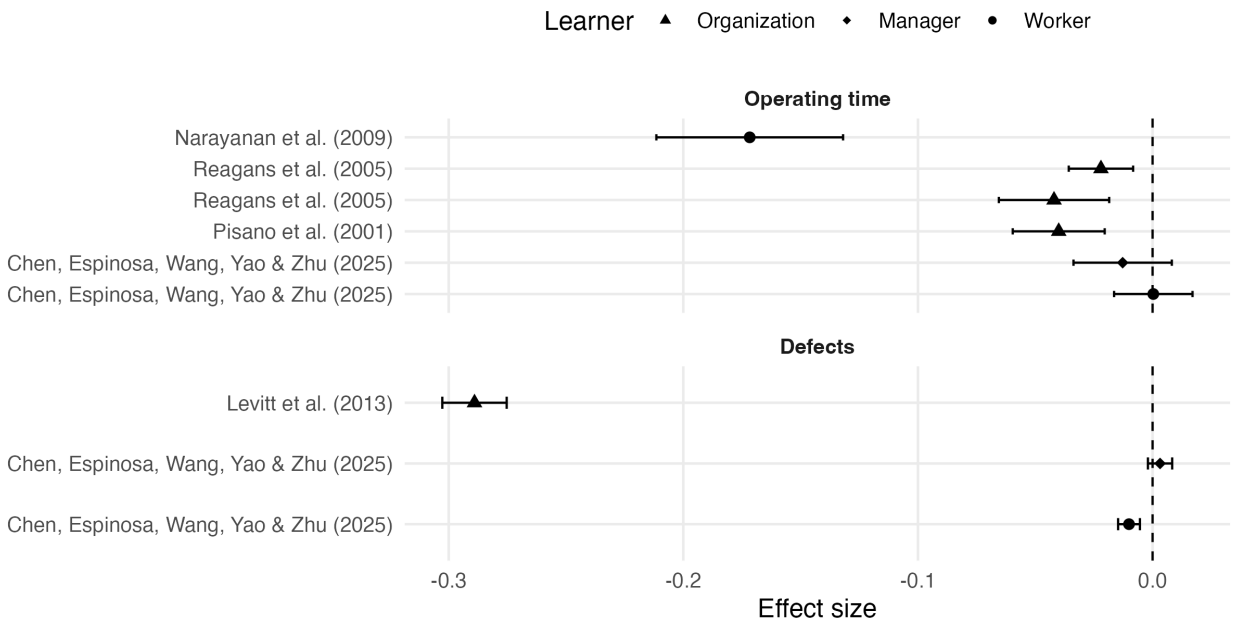
Dependent variable	Total number of models produced by the team (1)	(2)
Early avg. output (first two models)	9.754	
	(13.908)	
Early average defect rate (first two models)	-3.064	
	(212.493)	
Early avg. output (first three models)		5.610
		(10.991)
Early average defect rate (first three models)		96.786
		(242.496)
Observations	76	76
R-squared	0.007	0.005

Notes: This table reports team-level regressions of total production engagement on early team performance. The dependent variable is the total number of distinct models in which a team participates over the sample period. The sample is restricted to teams that participate in more than three models. Columns (1) and (2) measure early performance over each team's first two and first three completed models, respectively, ranked by completion time. Output is measured as the number of products within a model, and the defect rate as the share of products in a model with a defect. Standard errors are reported in parentheses.

C Supporting Materials for H1

C.1 Benchmarking Effect Sizes Against Prior Work

Figure A9: Comparison of Learning Effects Across Studies



Notes: This figure plots effect sizes and 95% confidence intervals from our study and selected prior work. Outcomes are grouped into operating time and defects. Shapes indicate the learning subject: triangles for organizations, diamonds for managers, and circles for workers. Effect sizes are reported as estimated coefficients on log (or IHS) cumulative experience.

C.2 Effects of Worker and Manager Experience When Estimated Separately

Table A7 re-estimates the specifications in Table 1 including only one type of experience (worker or manager) at a time. Because worker and manager experience accumulate together as a combination continues to produce the same model, they are positively correlated in the raw data ($r = 0.59$), so the included measure partly absorbs the effect of the omitted one.

In Panel A, worker experience alone appears to improve coordination outcomes: its co-

efficients on waiting time and material shortage both turn negative (from 0.014 and 0.003 in Table 1 to −0.013 and −0.003), and the material-shortage effect becomes marginally significant. In Panel B, manager experience alone acquires a negative coefficient on defects (from 0.003 to −0.003), while its coordination effects attenuate toward zero (waiting time from −0.039 to −0.030; material shortage from −0.009 to −0.007) as it absorbs part of the offsetting worker effect. A single-role specification therefore misattributes learning across roles, producing apparent effects on outcomes that the full specification assigns to the other role. Only when both types of experience enter simultaneously does the role-differentiated pattern emerge: worker experience improves execution quality, and manager experience improves coordination.

Table A7: Effects of Worker and Manager Experience (Separately Estimated)

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Only Worker Experience (Manager Experience Omitted)				
IHS(Exp. of Wkr. Combn.)	-0.009	-0.008	-0.013	-0.003
	(0.006)	(0.002)	(0.009)	(0.002)
Observations	76,690	76,690	76,690	76,690
R-squared	0.761	0.343	0.613	0.569
Panel B: Only Manager Experience (Worker Experience Omitted)				
IHS(Exp. of Mgr. Combn.)	-0.013	-0.003	-0.030	-0.007
	(0.008)	(0.002)	(0.012)	(0.002)
Observations	76,690	76,690	76,690	76,690
R-squared	0.761	0.342	0.614	0.570
Dependent Variable Mean	1.221	0.108	2.556	0.654
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table re-estimates the specifications in Table 1 including only one type of experience at a time, with the other omitted from the regression and therefore absorbed into the error term. Panel A includes only the IHS of worker-combination experience; Panel B includes only the IHS of manager-combination experience. The dependent variables, sample, fixed effects, and clustering are identical to Table 1: all regressions include fixed effects for product model, calendar month, and the specific worker and manager combinations involved in production, with standard errors clustered at the product model level and reported in parentheses.

C.3 Performance Outcomes and Aggregate Output

To verify that the improvements we documented in the intermediate production outcomes translate into higher overall productivity, we relate daily output to the four performance measures studied in the main analysis: operating time, waiting time, defects, and material shortages. We define daily output at the model-day level as the IHS-transformed number of products completed for a given model on a given day.[7] To align the intermediate outcomes with this unit of analysis, we average each measure across products completed for the focal model on that day. We then regress daily output on these four model-day measures, controlling for the same set of fixed effects.

Figure A10 shows that operating time (−0.073), waiting time (−0.071), and the defect rate (−0.226) are each negatively and significantly associated with daily output, whereas the material shortage rate is not (0.051). These results connect the intermediate performance measures to aggregate productivity: the reductions in defects and waiting time associated with worker and manager experience in our main analysis occur along margins that are also associated with higher daily output.

[7] We use the day as the temporal unit because production cycles are relatively short: 38% of products are completed within a single day, the median product takes 12 hours to complete, and 59.8% are completed within one week. Coarser temporal aggregation would therefore mask substantial variation in output.

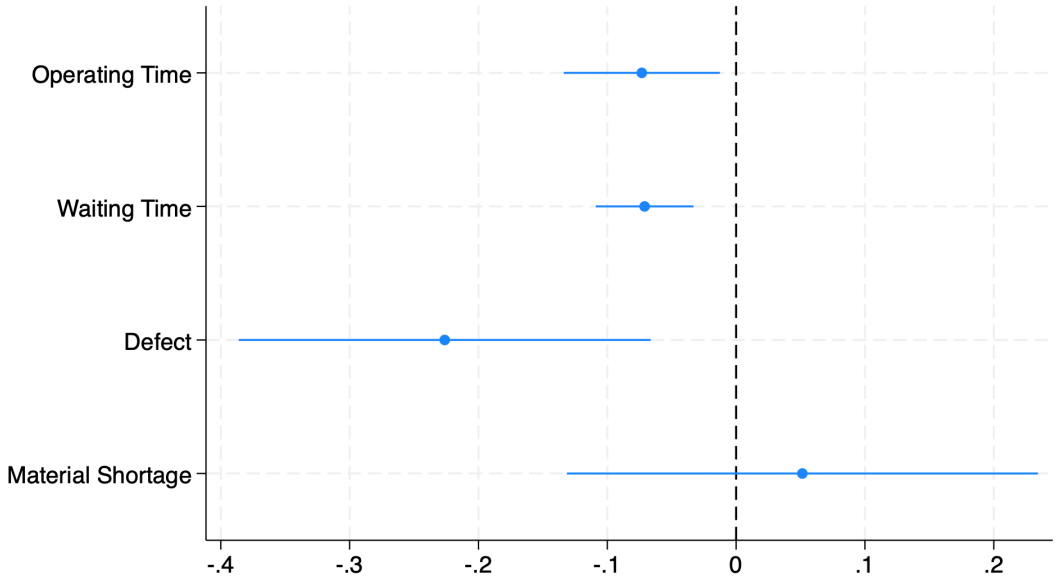


Figure A10: Effect of Production Outcomes on Output

Notes: The dependent variable is the IHS of daily output, defined as the number of products a model completes on a given day. Each independent variable is the model-day mean of the corresponding product-level measure, computed across the products completed on that model-day. Operating time and waiting time enter in inverse hyperbolic sine; defect and material shortage are model-day rates and enter in levels. All specifications control for the model-day mean of product workload, measured as the sum of the standard duration assigned by the firm to complete each task and entered in IHS, and absorb model, day, worker-combination, and manager-combination fixed effects, with standard errors clustered at the model level. Because the independent variables are defined over the products completed on a given model-day, the estimation sample consists of the 4,209 model-days on which at least one product is completed and for which the fixed effects are identified. Confidence intervals are shown at the 95% level.

D Supporting Materials for H2

D.1 Composite Measures of Managerial Intervention

As a robustness check, we construct two composite indices of coordination interventions, defined as the average and the minimum of the task-reordering and worker-reallocation rates. Table A8 re-estimates the specifications in Table 2 using these composite measures in place of the two individual intervention rates; both yield findings consistent with Table 2.

Table A8: Managerial Coordination Interventions and Learning Effects: Composite Intervention Indices

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Composite Intervention (Average of the Two Rates)				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.007	0.013	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
	[0.611]	[0.029]	[0.439]	[0.462]
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	0.002	-0.001
	(0.011)	(0.003)	(0.017)	(0.003)
	[0.397]	[0.397]	[0.906]	[0.906]
IHS(Exp. of Wkr. Combn.)× Composite Intervention	0.053	-0.024		
	(0.039)	(0.014)		
	[0.397]	[0.219]		
IHS(Exp. of Mgr. Combn.)× Composite Intervention			-0.307	-0.061
			(0.075)	(0.015)
			[0.000]	[0.000]
Observations	75,762	75,762	75,762	75,762
R-squared	0.752	0.332	0.607	0.567
Panel B: Composite Intervention (Minimum of the Two Rates)				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.007	0.013	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
	[0.606]	[0.039]	[0.443]	[0.466]
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	0.001	-0.001
	(0.011)	(0.003)	(0.017)	(0.003)
	[0.398]	[0.398]	[0.946]	[0.924]
IHS(Exp. of Wkr. Combn.)× Composite Intervention	0.055	-0.027		
	(0.040)	(0.014)		
	[0.398]	[0.168]		
IHS(Exp. of Mgr. Combn.)× Composite Intervention			-0.305	-0.062
			(0.077)	(0.016)
			[0.000]	[0.000]
Observations	75,762	75,762	75,762	75,762
R-squared	0.752	0.332	0.607	0.567
Dependent Variable Mean	1.221	0.108	2.556	0.654
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table reports the robustness check that interact worker and manager experience with a composite coordination-intervention index in place of the two individual intervention rates used in Table 2. Panel A constructs the composite as the average of the task-reordering and worker-reallocation rates; Panel B constructs it as their minimum. Both measures are model-level. All regressions include model, year-month, worker combination, and manager combination fixed effects. Standard errors, clustered at the product model level, are reported in parentheses. Benjamini–Hochberg FDR-adjusted q-values, computed jointly across the twelve experience-related coefficients within each panel (worker experience, manager experience, and the interaction across the four outcomes), are reported in brackets.

D.2 Ruling Out a Contemporaneous Response: A Backward-Looking Intervention Measure

One might be concerned that the rate of task reordering (or worker reallocation) is correlated with how quickly a manager is learning along the performance dimension, rather than reflecting a managerial behavior that complements experience: if managers who are already improving quickly also happen to reorder tasks more often, the interaction between managerial experience and intervention would capture this correlation rather than a genuine facilitating effect of intervention. To address this, we reconstruct each intervention measure as a backward-looking (running) rate, computed for each product using only the change indicators from products completed earlier in the same model by the same manager combination. Because this running measure reflects only adjustments that had already occurred by the time a given product was produced, it cannot mechanically pick up improvements or adjustments that happen later in the same manager's trajectory on this model.

Table A9 re-estimates the specifications in Table 2 using these running measures. Because a running measure is undefined for the first product handled by each manager combination within a model, the estimation sample is smaller (N=60,861 vs. 75,762 in Table 2). The interaction between managerial experience and intervention remains negative and significant for waiting time across both measures (p = 0.008 for task reordering; p = 0.005 for worker reallocation), roughly 40% of the magnitude obtained with the contemporaneous measure, while the interaction for material shortage is no longer distinguishable from zero.

Table A9: Managerial Coordination Interventions and Learning Effects: Backward-Looking Intervention Measure

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Task Reordering (Running)				
IHS(Exp. of Wkr. Combn.)	-0.006	-0.012	0.018	0.002
	(0.009)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.019	0.005	-0.042	-0.013
	(0.012)	(0.003)	(0.020)	(0.004)
IHS(Exp. of Wkr. Combn.)× Rate of Task Reordering (Running)	0.040	0.010		
	(0.035)	(0.009)		
IHS(Exp. of Mgr. Combn.)× Rate of Task Reordering (Running)			-0.121	-0.010
			(0.046)	(0.008)
Observations	60,861	60,861	60,861	60,861
R-squared	0.696	0.304	0.580	0.560
Panel B: Worker Reallocation (Running)				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.012	0.018	0.002
	(0.009)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.019	0.005	-0.041	-0.013
	(0.012)	(0.003)	(0.020)	(0.004)
IHS(Exp. of Wkr. Combn.)× Rate of Worker Reallocation (Running)	0.050	0.008		
	(0.035)	(0.009)		
IHS(Exp. of Mgr. Combn.)× Rate of Worker Reallocation (Running)			-0.130	-0.010
			(0.047)	(0.008)
Observations	60,861	60,861	60,861	60,861
R-squared	0.696	0.304	0.580	0.560
Dependent Variable Mean	1.221	0.108	2.556	0.654
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table re-estimates the specifications in Table 2 using a backward-looking (running) version of each intervention measure, addressing the concern that managerial intervention may respond contemporaneously to circumstances within the same model. For each product, the intervention rate is recomputed using only products completed prior to the focal product within the same manager combination on the same model, so that a product's intervention measure cannot mechanically reflect adjustments that occur later in the same model. Panel A interacts worker and manager experience with the running Rate of Task Reordering; Panel B with the running Rate of Worker Reallocation. Because a running measure is undefined for the first product handled by each manager combination within a model, the estimation sample (N=60,861) is smaller than in Table 2 (N=75,762). All regressions include model, year-month, worker combination, and manager combination fixed effects. Standard errors, clustered at the product model level, are reported in parentheses.

E Supporting Materials for H3

Table A10 reports the estimated complementarity between worker and manager experience based on Equation 4. The interaction between worker and manager experience is statistically significant for operating time but not after the FDR adjustment ($q = 0.111$). In contrast, the interaction remains statistically significant for waiting time after the adjustment ($q = 0.064$).

Table A10: How Managers' Experience Affect Workers' Learning

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
IHS(Exp. of Wkr. Combn.)	0.036	-0.009	0.081	0.009
	(0.021)	(0.005)	(0.033)	(0.006)
	[0.157]	[0.113]	[0.064]	[0.286]
IHS(Exp. of Mgr. Combn.)	-0.010	0.003	-0.034	-0.008
	(0.010)	(0.003)	(0.016)	(0.003)
	[0.352]	[0.288]	[0.105]	[0.064]
IHS(Exp. of Mgr. Combn.) × IHS(Exp. of Wkr. Combn.)	-0.009	-0.000	-0.018	-0.002
	(0.005)	(0.001)	(0.007)	(0.001)
	[0.111]	[0.732]	[0.064]	[0.352]
Dependent Variable Mean	1.221	0.108	2.556	0.654
Observations	76,690	76,690	76,690	76,690
R-squared	0.762	0.343	0.614	0.570
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: The dependent variables include the inverse hyperbolic sine (IHS) of total operating time, a binary indicator for whether a production defect occurred, the IHS of waiting time between tasks, and a binary indicator for whether a material shortage occurred. The key independent variables are the IHS of cumulative production experience for both the worker combination and the manager combination involved in producing the product, measured by the number of completed units prior to the current one. All regressions include fixed effects for product model, calendar month, and the specific worker and manager combinations involved in production. Standard errors, clustered at the product model level, are reported in parentheses; Benjamini-Hochberg FDR-adjusted q-values are reported in square brackets. FDR-adjusted q-values compute jointly across the twelve main experience-related coefficients in this table (worker experience, manager experience, and the interaction across the four outcomes).

F Supporting Materials for H4

As an alternative measure of managerial soft skills, we report the results using the factor-analysis-based index described in Section A.7. Table A11 reports the split-sample results for the complementarity between worker and manager experience under high and low managerial soft skills using the factor-analysis index, corresponding to the analysis using the PCA-based index in Section 4.4. Because the PCA- and factor-analysis-based indices are highly correlated (Spearman $\rho = 0.998$) and only 69 out of 64,102 observations receive different high/low classifications under the two approaches, the resulting estimates are nearly identical.

Table A11: Heterogeneity of Complementarity Effects: Managerial Soft Skills (Factor Analysis)

Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Operating Time)		Defect		IHS(Waiting Time)		Material Shortage	
	Managerial Soft Skills		Managerial Soft Skills		Managerial Soft Skills		Managerial Soft Skills	
	High	Low	High	Low	High	Low	High	Low
IHS(Exp. of Wkr. Combn.)	0.037	0.043	-0.004	-0.003	0.191	0.035	0.043	0.012
	(0.046)	(0.026)	(0.010)	(0.007)	(0.074)	(0.038)	(0.014)	(0.007)
IHS(Exp. of Mgr. Combn.)	-0.064	0.024	-0.008	0.005	-0.078	-0.034	-0.028	-0.008
	(0.027)	(0.012)	(0.006)	(0.004)	(0.042)	(0.020)	(0.008)	(0.004)
IHS(Exp. of Wkr. Combn.) × IHS(Exp. of Mgr. Combn.)	-0.007	-0.014	-0.002	-0.002	-0.031	-0.011	-0.006	-0.003
	(0.010)	(0.005)	(0.003)	(0.001)	(0.017)	(0.008)	(0.003)	(0.002)
P-value for equality of Interaction effects (High vs. Low subsamples)	0.524		0.868		0.296		0.474	
Dependent Variable Mean	1.969	1.049	0.144	0.104	2.930	2.559	0.561	0.650
Observations	15,625	45,340	15,625	45,340	15,625	45,340	15,625	45,340
R-squared	0.865	0.759	0.498	0.336	0.740	0.641	0.629	0.620
Year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables include the inverse hyperbolic sine (IHS) of total operating time, a binary indicator for whether a production defect occurred, the IHS of waiting time between tasks, and a binary indicator for whether a material shortage occurred. Columns (1)–(8) split the sample by a factor-analysis-based index of managerial soft skills constructed from workers' survey evaluations. The index summarizes five dimensions: responsiveness, problem diagnosis, problem-solving, understanding of workers, and communication. Standard errors are clustered at the product-model level.

G Robustness Checks

G.1 Experience Accumulated Outside the Focal Team–model Pair

A potential concern is that our estimates capture experience accumulated outside the focal team-model pair. We address this with two sets of controls. First, we control for members' individual familiarity with the same model, measured as the average cumulative model-specific experience of all members (workers and managers) involved in producing the focal unit. This measure is defined over each member's full history of assignments to the model regardless of team composition, and therefore includes, rather than nets out, any experience accumulated while working in the focal combination itself. Second, we control for the team combination's cumulative production volume across all models. A combination that has worked together on other models may perform better for reasons unrelated to model-specific experience, so this control captures general familiarity among members that is independent of the focal model. Together, the two controls help confirm that our estimates reflect the experience of the same members working together on the same model, rather than members' individual experience or their familiarity with one another in general. To ensure that our results are not driven by functional-form assumptions, each control enters flexibly, using alternative percentile-bin specifications that partition its distribution into three, five, and ten groups, together with a continuous measure of experience.

Figures A11, A13 and A15 control for members' individual model-specific experience, while Figures A12, A14 and A16 control for the combination's cumulative production volume across all models. For H1, the role-differentiated learning results remain stable under both sets of controls, except that the effect of manager-combination experience on waiting time becomes insignificant under the finer percentile partitions of individual model-specific experience. This likely reflects the relative stability of manager combinations: when combination membership rarely changes, this individual-level control becomes mechanically close to the combination-specific experience measure, reducing precision when the control is en-

tered more flexibly. For H2, the managerial-intervention results are also broadly stable under both sets of controls. For both intervention rates, the coordination interaction with manager experience remains negative and significant across nearly all specifications, while the execution interaction with worker experience remains only marginally significant and loses significance under the finest percentile partitions of individual model-specific experience. For H3, the waiting-time interaction remains stable across specifications, becoming insignificant only when individual model-specific experience enters continuously. The operating-time interaction remains negative across both sets of controls, although its statistical significance varies across specifications.

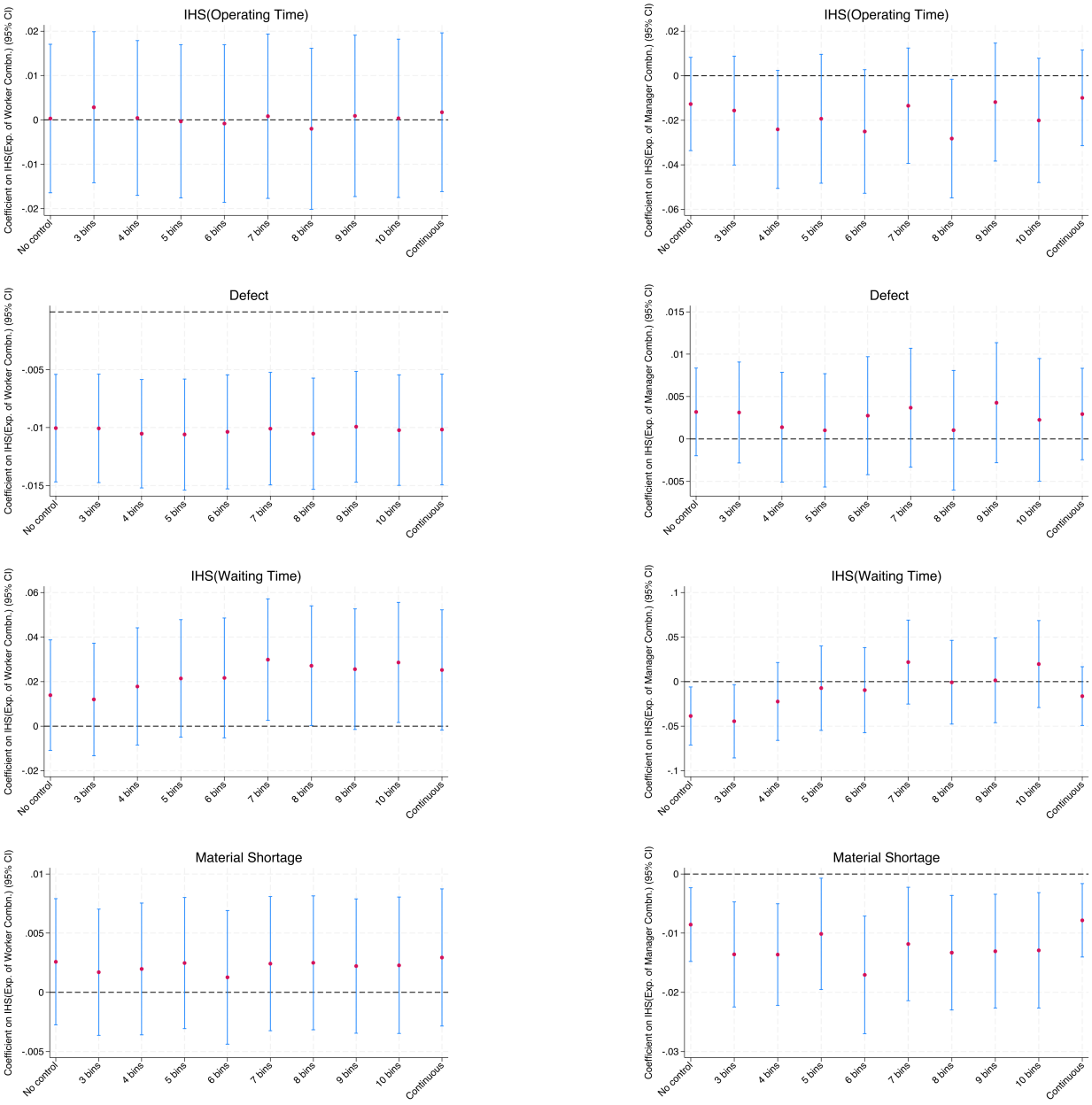


Figure A11: Effects Across Specifications Controlling for Individual Model-Specific Experience

Notes: This figure reports coefficient estimates and 95% confidence intervals for worker- and manager-combination experience controlling for members' individual model-specific experience. "No control" refers to the baseline specification without this additional control. The remaining specifications control for the average cumulative model-specific experience of all members (workers and managers) involved in producing the focal unit, accumulated across their prior assignments regardless of team composition, using percentile-bin specifications that partition the control variable into three through ten groups, as well as a continuous specification based on its inverse hyperbolic sine transformation. Red dots indicate point estimates and blue bars indicate 95% confidence intervals.

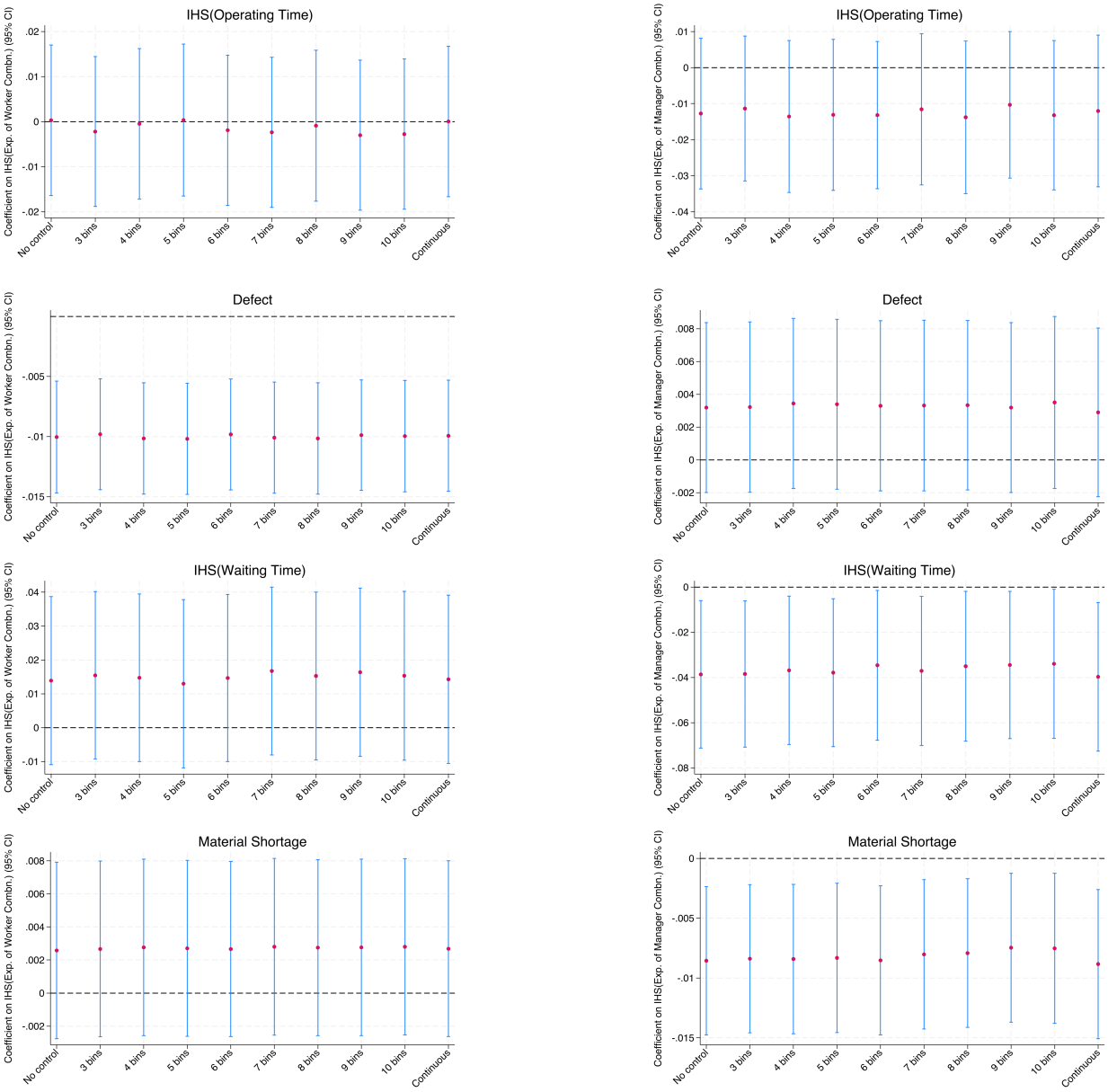


Figure A12: Effects Across Specifications Controlling for the Combination's Cumulative Production Volume Across All Models

Notes: This figure reports coefficient estimates and 95% confidence intervals for worker- and manager-combination experience controlling for the team combination's general co-production experience. "No control" refers to the baseline specification without this additional control. The remaining specifications control for the team combination's cumulative production volume, measured as the total number of products the combination produced across all models prior to the current product, using percentile-bin specifications that partition the control variable into three through ten groups, as well as a continuous specification using the raw value. Red dots indicate point estimates and blue bars indicate 95% confidence intervals.

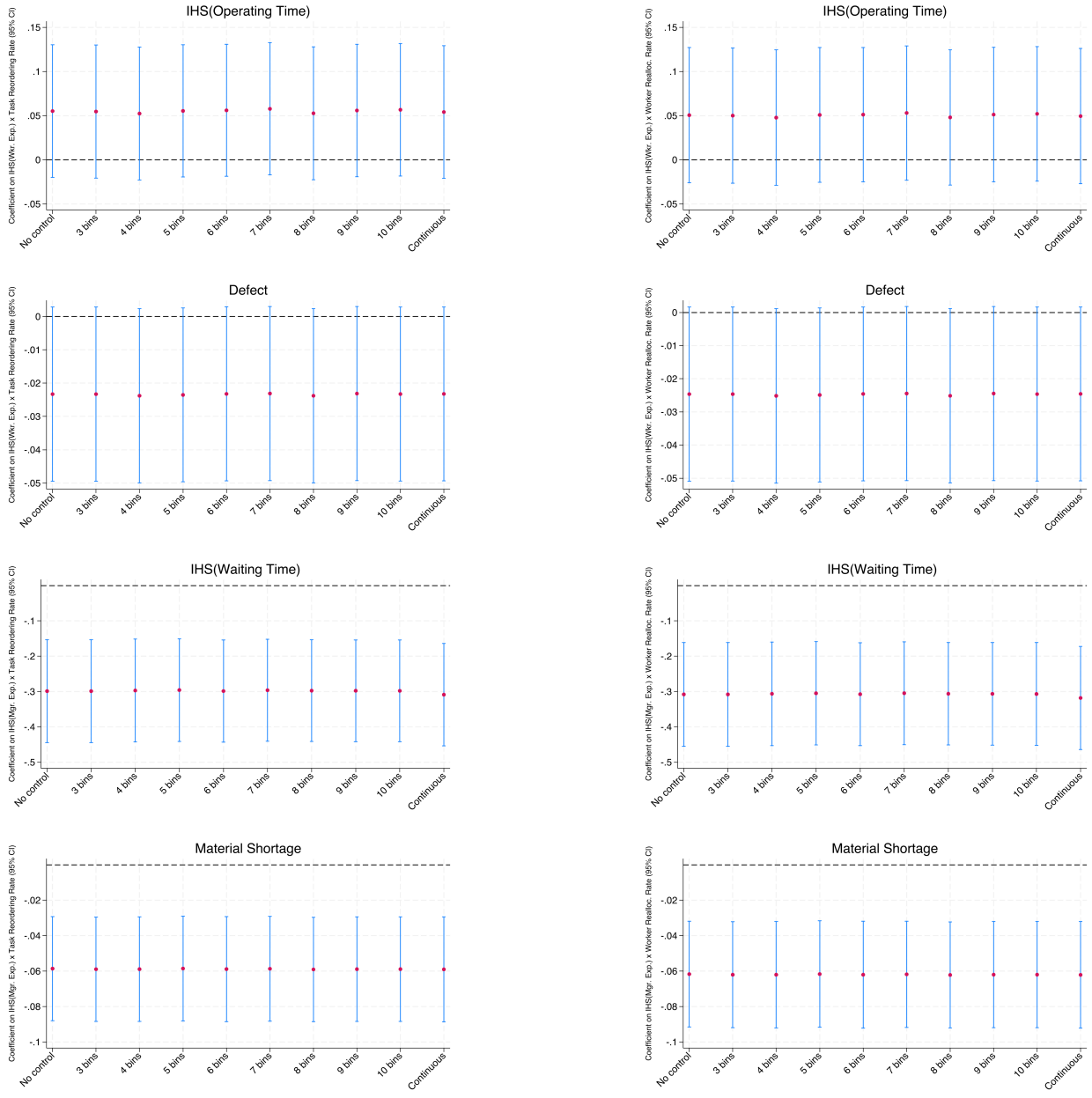


Figure A13: Effects of Managerial Intervention Across Specifications Controlling for Individual Model-Specific Experience

Notes: This figure repeats the exercise in Figure A11 for the managerial-intervention specification in Table 2. Each row corresponds to one outcome; the left column interacts experience with the Rate of Task Reordering and the right column with the Rate of Worker Reallocation. Because the specification is asymmetric, the coefficient plotted for execution outcomes (operating time, defect) is the interaction with worker experience, and for coordination outcomes (waiting time, material shortage) the interaction with manager experience. "No control" refers to the baseline specification without the additional control. The remaining specifications control for the average cumulative model-specific experience of all members involved in producing the focal unit, using percentile-bin specifications that partition the control variable into three through ten groups, as well as a continuous specification. Red dots indicate point estimates and blue bars indicate 95% confidence intervals.

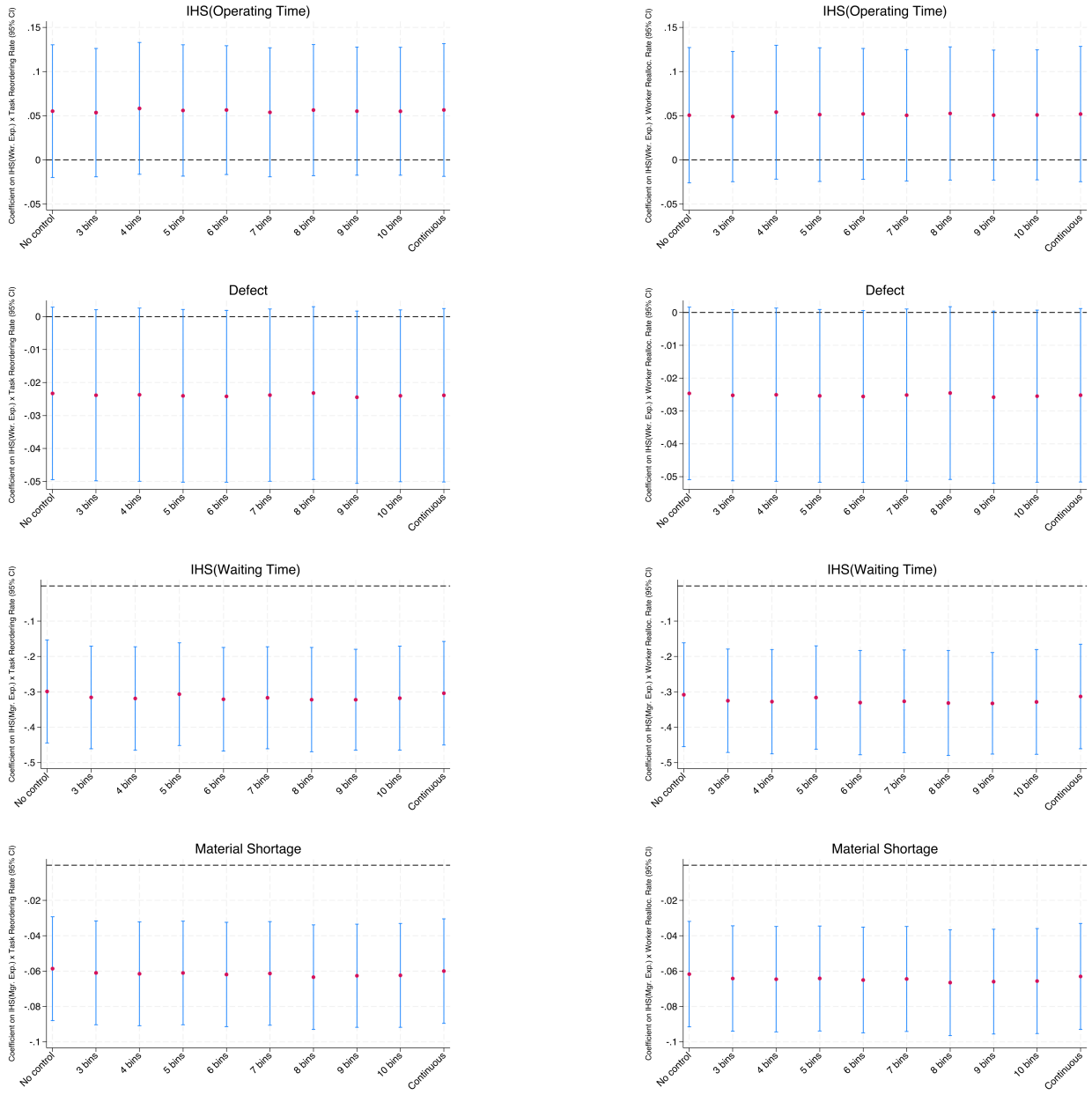


Figure A14: Effects of Managerial Intervention Across Specifications Controlling for the Combination's Cumulative Production Volume Across All Models

Notes: This figure repeats the exercise in Figure A13 controlling for the team combination's cumulative production volume across all models instead of individual model-specific experience. See the notes to Figure A13 for the definition of the plotted coefficient. Red dots indicate point estimates and blue bars indicate 95% confidence intervals.

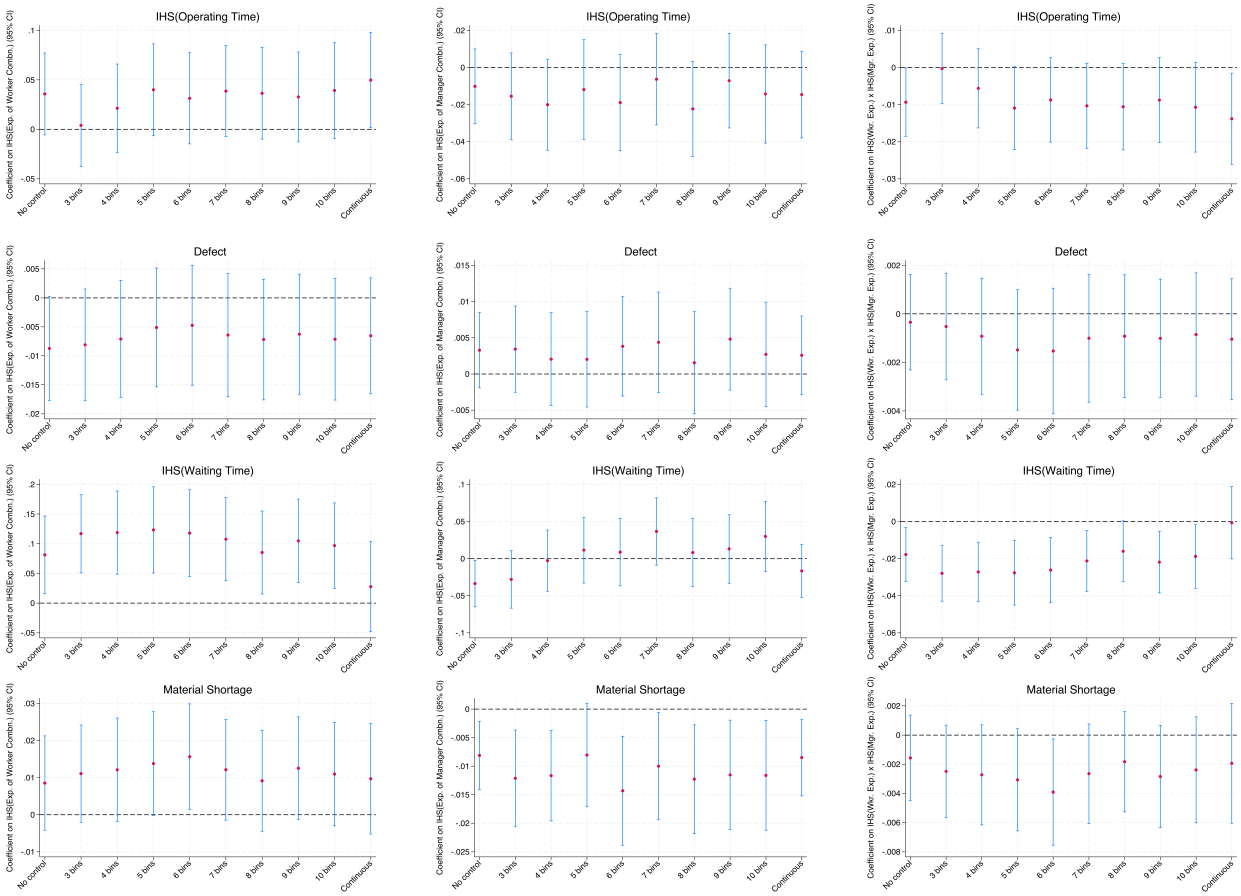


Figure A15: Complementarity Across Specifications Controlling for Individual Model-Specific Experience

Notes: This figure repeats the exercise in Figure A11 for the specification that includes the interaction between worker- and manager-combination experience. Each row corresponds to one outcome, and the three columns report, from the same regression, the coefficients on worker-combination experience, manager-combination experience, and their interaction. In the presence of the interaction, the coefficients in the first two columns are marginal effects evaluated at zero experience of the other role. "No control" refers to the baseline specification without the additional control. The remaining specifications control for the average cumulative model-specific experience of all members involved in producing the focal unit, using percentile-bin specifications that partition the control variable into three through ten groups, as well as a continuous specification based on its inverse hyperbolic sine transformation. Red dots indicate point estimates and blue bars indicate 95% confidence intervals.

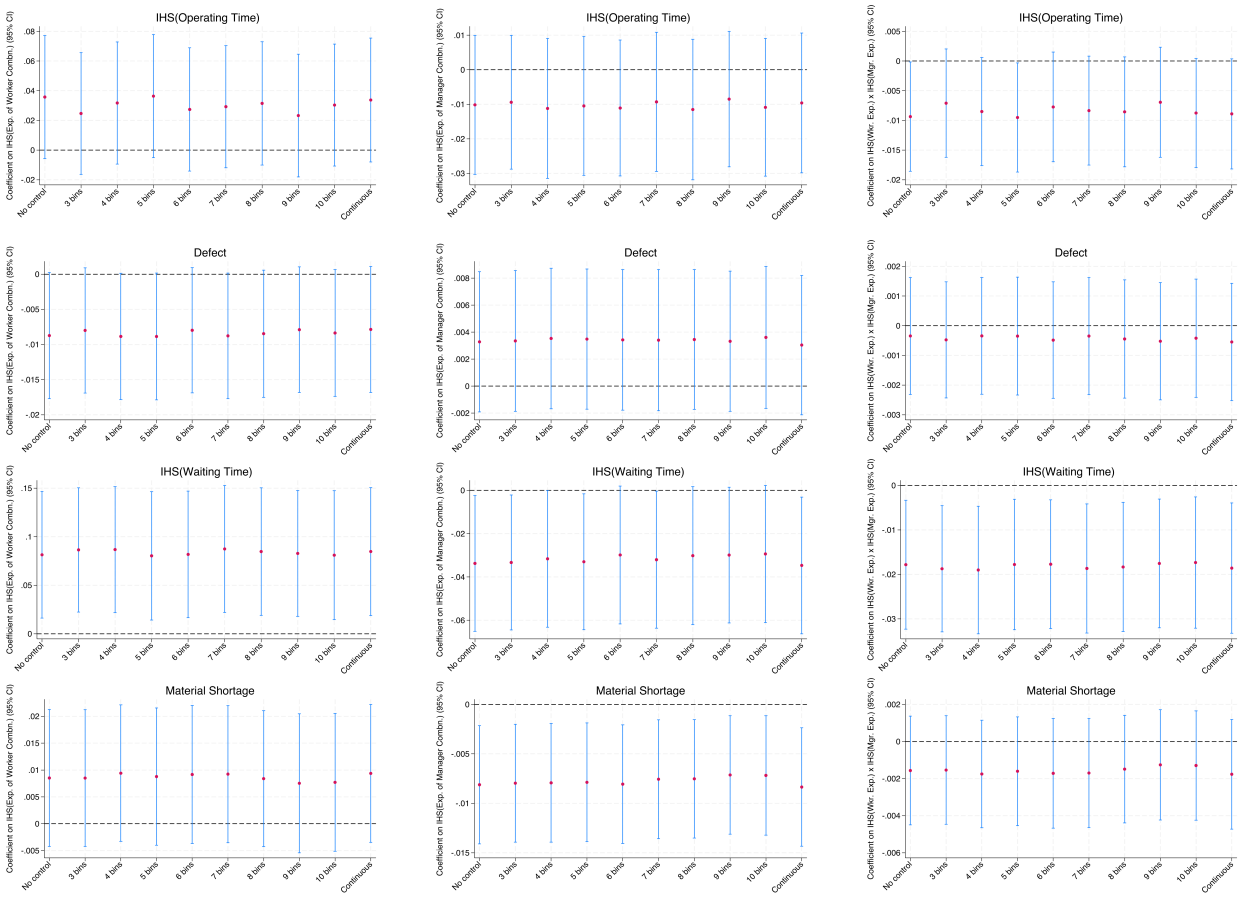


Figure A16: Complementarity Across Specifications Controlling for the Combination's Cumulative Production Volume Across All Models

Notes: This figure repeats the exercise in Figure A12 for the specification that includes the interaction between worker- and manager-combination experience. Each row corresponds to one outcome, and the three columns report, from the same regression, the coefficients on worker-combination experience, manager-combination experience, and their interaction. In the presence of the interaction, the coefficients in the first two columns are marginal effects evaluated at zero experience of the other role. "No control" refers to the baseline specification without the additional control. The remaining specifications control for the team combination's cumulative production volume across all models prior to the current product, using percentile-bin specifications that partition the control variable into three through ten groups, as well as a continuous specification using the raw value. Red dots indicate point estimates and blue bars indicate 95% confidence intervals.

G.2 Endogenous Team Composition

Another concern is that team composition may be endogenously determined. For example, poorly functioning teams may be more likely to dissolve, while more effective teams may remain together longer. Such dynamics could bias our estimates by jointly affecting experience accumulation and performance outcomes.

To address this concern, we control for a time-varying measure of team instability. For each manager combination in each month, we compute the share of workers under those managers who left the firm in the prior month and include this lagged exit rate as a control variable. This absorbs time-varying instability that may simultaneously affect turnover and production outcomes.

The results are reported in Tables A12–A15. Within each table, Panels A–C correspond to the three checks described above: controlling for cumulative cross-model collaboration experience, restricting the sample to worker–manager combinations that complete at least two products together within the same model, and controlling for the lagged monthly worker exit rate, respectively. Across all three checks, the main findings remain robust. The role-differentiated learning patterns, the facilitating effects of managerial intervention, and the complementarity in learning trajectories are qualitatively unchanged. These results suggest that our findings are not driven by selective team dissolution or time-varying team instability.

Table A12: Effects of Worker and Manager Experience on Performance: Robustness to Team Composition and Turnover

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Controlling for Worker–Manager Combination Stability				
IHS(Exp. of Wkr. Combn.)	-0.001	-0.009	0.019	0.003
	(0.009)	(0.002)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	-0.038	-0.008
	(0.011)	(0.003)	(0.017)	(0.003)
Observations	76,690	76,690	76,690	76,690
Panel B: Restricting to Stable Worker-Manager Pairs				
IHS(Exp. of Wkr. Combn.)	-0.001	-0.010	0.011	0.002
	(0.009)	(0.002)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.012	0.003	-0.038	-0.008
	(0.011)	(0.003)	(0.018)	(0.003)
Observations	71,352	71,352	71,352	71,352
Panel C: Controlling for Team Exit Rate				
IHS(Exp. of Wkr. Combn.)	0.000	-0.010	0.014	0.003
	(0.009)	(0.002)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	-0.039	-0.009
	(0.011)	(0.003)	(0.017)	(0.003)
Observations	76,598	76,598	76,598	76,598
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: The dependent variables are the IHS of total operating time, a defect indicator, the IHS of waiting time, and a material shortage indicator. The key independent variables are the IHS of cumulative experience for the worker and manager combinations involved in producing each product. Panel A additionally controls for cumulative cross-model collaboration experience between worker and manager combinations. Panel B restricts the sample to worker-manager pairs collaborating across at least three distinct product models. Panel C controls for the lagged monthly exit rate of workers under the managers in each combination, defined as the share of workers who left the firm in the prior month. All regressions include fixed effects for product model, calendar month, and worker and manager combinations. Standard errors (in parentheses) are clustered at the product model level.

Table A13: Managerial Coordination Interventions and Learning Effects (Rate of Task Reordering): Robustness to Team Composition and Turnover

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Controlling for Worker–Manager Combination Stability				
IHS(Exp. of Wkr. Combn.)	-0.008	-0.006	0.019	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.013	0.004	0.002	-0.001
	(0.011)	(0.003)	(0.017)	(0.003)
IHS(Exp. of Wkr. Combn.)× Rate of Task Reordering	0.056	-0.024		
	(0.038)	(0.013)		
IHS(Exp. of Mgr. Combn.)× Rate of Task Reordering			-0.302	-0.059
			(0.074)	(0.015)
Dependent Variable Mean	1.221	0.108	2.556	0.654
Observations	75,762	75,762	75,762	75,762
R-squared	0.752	0.332	0.607	0.567
Panel B: Restricting to Stable Worker–Manager Pairs				
IHS(Exp. of Wkr. Combn.)	-0.008	-0.006	0.010	0.002
	(0.010)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.012	0.003	0.001	0.000
	(0.011)	(0.003)	(0.019)	(0.003)
IHS(Exp. of Wkr. Combn.)× Rate of Task Reordering	0.053	-0.033		
	(0.040)	(0.014)		
IHS(Exp. of Mgr. Combn.)× Rate of Task Reordering			-0.291	-0.061
			(0.081)	(0.016)
Dependent Variable Mean	1.087	0.0985	2.453	0.656
Observations	71,352	71,352	71,352	71,352
R-squared	0.747	0.332	0.609	0.564
Panel C: Controlling for Team Exit Rate				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.007	0.013	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	0.001	-0.001
	(0.011)	(0.003)	(0.017)	(0.003)
IHS(Exp. of Wkr. Combn.)× Rate of Task Reordering	0.055	-0.023		
	(0.039)	(0.013)		
IHS(Exp. of Mgr. Combn.)× Rate of Task Reordering			-0.301	-0.059
			(0.074)	(0.015)
Dependent Variable Mean	1.221	0.108	2.556	0.654
Observations	75,674	75,674	75,674	75,674
R-squared	0.752	0.332	0.606	0.566
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table extends the team-composition robustness check to the managerial-intervention specification in Table 2, interacting worker and manager experience with the Rate of Task Reordering. Panel A controls for cumulative cross-model collaboration experience between worker and manager combinations. Panel B restricts the sample to worker–manager pairs that complete at least two products together within the same model. Panel C controls for the lagged monthly exit rate of workers under the managers in each combination, defined as the share of workers who left the firm in the prior month. All regressions include fixed effects for product model, calendar month, and worker and manager combinations. Standard errors (in parentheses) are clustered at the product model level.

Table A14: Managerial Coordination Interventions and Learning Effects (Rate of Worker Reallocation): Robustness to Team Composition and Turnover

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Controlling for Worker–Manager Combination Stability				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.006	0.019	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.013	0.004	0.004	-0.000
	(0.011)	(0.003)	(0.017)	(0.003)
IHS(Exp. of Wkr. Combn.)× Rate of Worker Reallocation	0.052	-0.026		
	(0.039)	(0.013)		
IHS(Exp. of Mgr. Combn.)× Rate of Worker Reallocation			-0.312	-0.062
			(0.075)	(0.015)
Dependent Variable Mean	1.221	0.108	2.556	0.654
Observations	75,762	75,762	75,762	75,762
R-squared	0.752	0.332	0.607	0.567
Panel B: Restricting to Stable Worker–Manager Pairs				
IHS(Exp. of Wkr. Combn.)	-0.007	-0.005	0.010	0.002
	(0.011)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.012	0.003	0.002	0.001
	(0.011)	(0.003)	(0.019)	(0.003)
IHS(Exp. of Wkr. Combn.)× Rate of Worker Reallocation	0.046	-0.034		
	(0.041)	(0.014)		
IHS(Exp. of Mgr. Combn.)× Rate of Worker Reallocation			-0.301	-0.064
			(0.082)	(0.016)
Dependent Variable Mean	1.087	0.0985	2.453	0.656
Observations	71,352	71,352	71,352	71,352
R-squared	0.747	0.332	0.609	0.564
Panel C: Controlling for Team Exit Rate				
IHS(Exp. of Wkr. Combn.)	-0.006	-0.007	0.013	0.003
	(0.010)	(0.003)	(0.013)	(0.003)
IHS(Exp. of Mgr. Combn.)	-0.013	0.003	0.002	-0.000
	(0.011)	(0.003)	(0.017)	(0.003)
IHS(Exp. of Wkr. Combn.)× Rate of Worker Reallocation	0.050	-0.025		
	(0.039)	(0.013)		
IHS(Exp. of Mgr. Combn.)× Rate of Worker Reallocation			-0.310	-0.062
			(0.075)	(0.015)
Dependent Variable Mean	1.221	0.108	2.556	0.654
Observations	75,674	75,674	75,674	75,674
R-squared	0.752	0.332	0.607	0.566
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table extends the team-composition robustness check to the managerial-intervention specification in Table 2, interacting worker and manager experience with the Rate of Worker Reallocation. Panel A controls for cumulative cross-model collaboration experience between worker and manager combinations. Panel B restricts the sample to worker–manager pairs that complete at least two products together within the same model. Panel C controls for the lagged monthly exit rate of workers under the managers in each combination, defined as the share of workers who left the firm in the prior month. All regressions include fixed effects for product model, calendar month, and worker and manager combinations. Standard errors (in parentheses) are clustered at the product model level.

Table A15: Interaction Effects of Worker and Manager Experience: Robustness to Team Composition and Turnover

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	IHS(Operating Time)	Defect	IHS(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Controlling for Worker–Manager Combination Stability				
IHS(Exp. of Wkr. Combn.)	0.035	-0.008	0.085	0.009
	(0.021)	(0.005)	(0.033)	(0.006)
IHS(Exp. of Mgr. Combn.)	-0.010	0.003	-0.033	-0.008
	(0.010)	(0.003)	(0.016)	(0.003)
IHS(Exp. of Wkr. Combn.) × IHS(Exp. of Mgr. Combn.)	-0.009	-0.000	-0.018	-0.002
	(0.005)	(0.001)	(0.007)	(0.001)
Observations	76,690	76,690	76,690	76,690
Panel B: Restricting to Stable Worker–Manager Combinations				
IHS(Exp. of Wkr. Combn.)	0.033	-0.008	0.077	0.009
	(0.022)	(0.005)	(0.034)	(0.007)
IHS(Exp. of Mgr. Combn.)	-0.009	0.003	-0.032	-0.007
	(0.011)	(0.003)	(0.017)	(0.003)
IHS(Exp. of Wkr. Combn.) × IHS(Exp. of Mgr. Combn.)	-0.009	-0.000	-0.017	-0.002
	(0.005)	(0.001)	(0.008)	(0.001)
Observations	71,352	71,352	71,352	71,352
Panel C: Controlling for Team Exit Rate				
IHS(Exp. of Wkr. Combn.)	0.036	-0.009	0.081	0.009
	(0.021)	(0.005)	(0.033)	(0.007)
IHS(Exp. of Mgr. Combn.)	-0.010	0.003	-0.034	-0.008
	(0.010)	(0.003)	(0.016)	(0.003)
IHS(Exp. of Wkr. Combn.) × IHS(Exp. of Mgr. Combn.)	-0.009	-0.000	-0.018	-0.002
	(0.005)	(0.001)	(0.007)	(0.001)
Observations	76,598	76,598	76,598	76,598
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: The dependent variables are the IHS of total operating time, a defect indicator, the IHS of waiting time, and a material shortage indicator. The key independent variables are the IHS of cumulative experience for the worker and manager combinations involved in producing each product. High manager experience is defined as being above the median level of manager experience within a given product model. Panel A additionally controls for cumulative cross-model collaboration experience between worker and manager combinations. Panel B restricts the sample to worker–manager combinations collaborating across at least three distinct product models. Panel C controls for the lagged monthly exit rate of workers under the managers in each combination, defined as the share of workers who left the firm in the prior month. All regressions include fixed effects for product model, calendar month, and worker and manager combinations. Standard errors (in parentheses) are clustered at the product model level.

G.3 Alternative Functional Form: Logarithm Transformation

We next examine whether our results are sensitive to the choice of functional form. In our baseline specifications, we use the inverse hyperbolic sine (IHS) transformation to accommodate zero-valued and skewed outcomes. As a robustness check, we re-estimate all regressions using log-transformed outcomes, excluding waiting-time observations with non-positive values (about 10% of the sample) that arise from task overlaps. The results, reported in Tables A16–A18, are qualitatively similar to our baseline IHS estimates, confirming that our findings are not driven by the choice of transformation.

Table A16: Effects of Worker and Manager Experience on Performance (Logarithm Transformation)

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	Log(Operating Time)	Defect	Log(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Log(Exp. of Wkr. Combn.)	-0.000	-0.012	-0.001	0.003
	(0.008)	(0.003)	(0.012)	(0.003)
Log(Exp. of Mgr. Combn.)	-0.012	0.004	-0.033	-0.010
	(0.010)	(0.003)	(0.013)	(0.004)
P-value (H0: effect of worker = effect of manager)	0.488	0.004	0.176	0.060
Dependent Variable Mean	0.990	0.108	2.306	0.654
Observations	76,690	76,690	73,681	76,690
R-squared	0.769	0.343	0.655	0.570
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table extends the functional-form robustness check in Appendix G.3 to the role-differentiated specification in Table 1, re-estimating the specification using the natural logarithm, $\log(x+1)$, in place of the inverse hyperbolic sine transformation for experience, workload, operating time, and waiting time. This transformation is undefined only for the small share of observations with waiting time of one hour or more below zero, which arise from overlapping tasks; these (about 3.9% of the sample) are excluded, so Column (3) has a smaller sample than the other columns, so Column (3) has a smaller sample than the other columns, as in Tables A17 and A18 below. All regressions include model, year-month, worker combination, and manager combination fixed effects. Standard errors, clustered at the product model level, are reported in parentheses.

Table A17: Managerial Coordination Interventions and Learning Effects: Logarithmic Transformation

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	Log(Operating Time)	Defect	Log(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Panel A: Task Reordering				
Log(Exp. of Wkr. Combn.)	-0.006	-0.008	-0.001	0.003
	(0.010)	(0.003)	(0.012)	(0.003)
Log(Exp. of Mgr. Combn.)	-0.012	0.004	-0.002	-0.000
	(0.010)	(0.003)	(0.014)	(0.004)
Log(Exp. of Wkr. Combn.)× Rate of Task Reordering	0.048	-0.031		
	(0.039)	(0.017)		
Log(Exp. of Mgr. Combn.)× Rate of Task Reordering			-0.241	-0.073
			(0.057)	(0.019)
Dependent Variable Mean	0.990	0.108	2.306	0.654
Observations	75,762	75,762	72,838	75,762
R-squared	0.760	0.332	0.648	0.567
Panel B: Worker Reallocation				
Log(Exp. of Wkr. Combn.)	-0.006	-0.008	-0.001	0.003
	(0.010)	(0.003)	(0.012)	(0.003)
Log(Exp. of Mgr. Combn.)	-0.012	0.004	-0.000	0.000
	(0.010)	(0.003)	(0.014)	(0.004)
Log(Exp. of Wkr. Combn.)× Rate of Worker Reallocation	0.043	-0.033		
	(0.040)	(0.017)		
Log(Exp. of Mgr. Combn.)× Rate of Worker Reallocation			-0.255	-0.077
			(0.058)	(0.019)
Dependent Variable Mean	0.990	0.108	2.306	0.654
Observations	75,762	75,762	72,838	75,762
R-squared	0.760	0.332	0.648	0.567
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table extends the functional-form robustness check in Appendix G.3 to the managerial-intervention specification in Table 2, re-estimating the specification using the natural logarithm, $\log(x+1)$, in place of the inverse hyperbolic sine transformation for experience, workload, operating time, and waiting time. Panel A interacts worker and manager experience with the Rate of Task Reordering; Panel B with the Rate of Worker Reallocation. All regressions include model, year-month, worker combination, and manager combination fixed effects. Standard errors, clustered at the product model level, are reported in parentheses.

Table A18: How Managers' Experience Affect Workers' Learning (Logarithm Transformation)

Dependent Variables	Execution-based Outcomes		Coordination-based Outcomes	
	Log(Operating Time)	Defect	Log(Waiting Time)	Material Shortage
	(1)	(2)	(3)	(4)
Log(Exp. of Wkr. Combn.)	0.029	-0.011	0.047	0.009
	(0.019)	(0.005)	(0.027)	(0.008)
Log(Exp. of Mgr. Combn.)	-0.010	0.004	-0.028	-0.009
	(0.010)	(0.003)	(0.013)	(0.004)
Log(Exp. of Wkr. Combn.) × Log(Exp. of Mgr. Combn.)	-0.009	-0.000	-0.015	-0.002
	(0.005)	(0.001)	(0.007)	(0.002)
Dependent Variable Mean	0.990	0.108	2.306	0.654
Observations	76,690	76,690	73,681	76,690
R-squared	0.769	0.343	0.655	0.570
Year-month FE	Yes	Yes	Yes	Yes
Model FE	Yes	Yes	Yes	Yes
Worker Combination FE	Yes	Yes	Yes	Yes
Manager Combination FE	Yes	Yes	Yes	Yes

Notes: This table extends the functional-form robustness check in Appendix G.3 to the complementarity specification in Table A10, re-estimating the specification using the natural logarithm, $\log(x+1)$, in place of the inverse hyperbolic sine transformation for experience, workload, operating time, and waiting time. Log(Exp. of Wkr. Combn.) × Log(Exp. of Mgr. Combn.) captures the complementarity between worker and manager experience. All regressions include model, year-month, worker combination, and manager combination fixed effects. Standard errors, clustered at the product model level, are reported in parentheses.

H Comparison with Prior Work

We conducted a structured search of peer-reviewed studies on organizational learning spanning management and economics. We began with foundational reviews and classic work (e.g., Argote, 2012; Argote et al., 2021; Epple et al., 1996) and then used a combination of keyword searches and backward/forward citation tracking. Keyword searches included combinations of organizational learning, learning-by-doing, learning curve, experience, team experience, shared experience, manager experience in databases such as Google Scholar.

We present these papers in Table A19. The table is split into panels based on type (5 theoretical studies and 41 empirical studies) and on the level of empirical analysis (firm/team/individual). There are also multi-level studies, for which experience is measured at different levels spanning firm, team and individual. For each paper, we code (i) who the learners are, (ii) the performance outcome studied, (iii) how experience is measured, and (iv) the core theoretical prediction and implication. Figure A17 summarizes these papers in a literature map. Table A1 shows the dependent variables and Figure A9 plots the effect sizes in our paper and in prior works.

There are several patterns that are salient in Table A19 and Figure A17. First, a large literature documents experience–performance relationships at the firm or plant level (e.g., Benkard, 2000; Epple et al., 1996; Levitt et al., 2013; Pisano et al., 2001). While these studies establish robust aggregate learning curves, they often leave the microfoundations of learning underspecified: *which actors convert experience into improvement*, and *through which channels*? Recently, multilevel work shows that experience at different levels (individual/team/organization) can operate through distinct mechanisms (e.g., Chen, 2021; Espinosa et al., 2007; Fong Boh et al., 2007; Huckman et al., 2009; Jain, 2013; Reagans et al., 2005; Taylor and Greve, 2006), but it typically does not distinguish *formal hierarchical roles within teams* or identify role-specific learning mechanisms. We distinguish the learning of workers and managers within the same team.

Second, existing studies vary in both the dependent variables and the context (institu-

tions, settings, etc.), making comparisons difficult. As Harvey et al. (2022) note, "despite sharing the same label, performance outcomes predicted by team learning in one study (e.g., sales teams) do not necessarily equate to those predicted in another (e.g., new product development teams)." We are not aware of any study that examines multiple performance dimensions within the same context/organization. While some studies consider multiple outcomes, these outcomes essentially represent one dimension of performance (Alcacer & Oxley, 2014; Criaco et al., 2022; Hoang & Rothaermel, 2010; Huckman et al., 2009; Taylor & Greve, 2006; Weigelt, 2009). For example, Huckman et al. (2009) use defects and adherence (to schedule and effort estimates) as dependent variables, which are both measures of task quality. However, the five outcome variables in our study map into theoretically distinctive dimensions of performance: execution, coordination and ultimately, productivity.

Third, interaction effects have been examined across experience measured at different levels[8] and accumulated during different periods[9]. What remains under-tested is whether there are *cross-hierarchy* complementarities in hierarchical production teams and how such complementarities differ across performance dimensions.[10] Our results suggest that experience does not operate in isolation within hierarchical teams. Managers' experience can help workers translate their task experience into faster execution, while workers' experience can help managers translate their coordination experience into smoother workflow.

Fourth, existing studies test how the characteristics of tasks and teams modulate team's learning. These include complexity of task and complexity of coordination (Espinosa et al.,

[8]e.g., Jain (2013) tests complementarities between individual/team/firm experience, and Espinosa et al. (2007) test complementarities between task familiarity (individual experience) and team familiarity (shared team experience).

[9]e.g., Kor (2003) finds conflict effects between founder-based experience and shared team experience, and Zheng et al. (2016) find conflict effects between shared prior experience and current shared operating experience.

[10]A closely related study is Crocker and Eckardt (2014), which posits that individual human capital relates positively to individual performance and managerial unit-level HC weakens this link. Their logic is that high-quality managerial unit can raise the performance of low-HC individuals via better deployment, making individual HC less predictive of performance differences. However, their focus is human capital (knowledge/skills), and human capital comes from education, on-the-job training and experience accumulation. We directly focus on experience, a microfoundation of firms' dynamic capabilities. In contrast to them, we posit that within the same team, worker and manager experience reinforce each other because their roles (execution vs. coordination) are interdependent.

2007), the degree of task specialization (Schilling et al., 2003), task similarity or mismatch (Zheng, 2012; Zheng et al., 2016), diversity of members (Criaco et al., 2022) and intra-team trust (Zheng, 2012). Theoretical work emphasizes that managerial cognition and psychology (Helfat & Peteraf, 2015; Hodgkinson & Healey, 2011), managerial human and social capital (Helfat & Martin, 2015) and managerial design decisions (Harvey et al., 2022) can affect organizational learning, but these are not empirically tested. To our best knowledge, we are the first to test the modulating effects of managerial actions (task reordering and worker reallocation) and managerial soft skills on team learning. Even though Adhvaryu et al. (2023) shows certain managerial traits accelerate workers' learning, we show managerial skills can affect the learning of *both* workers and managers instead of workers only.

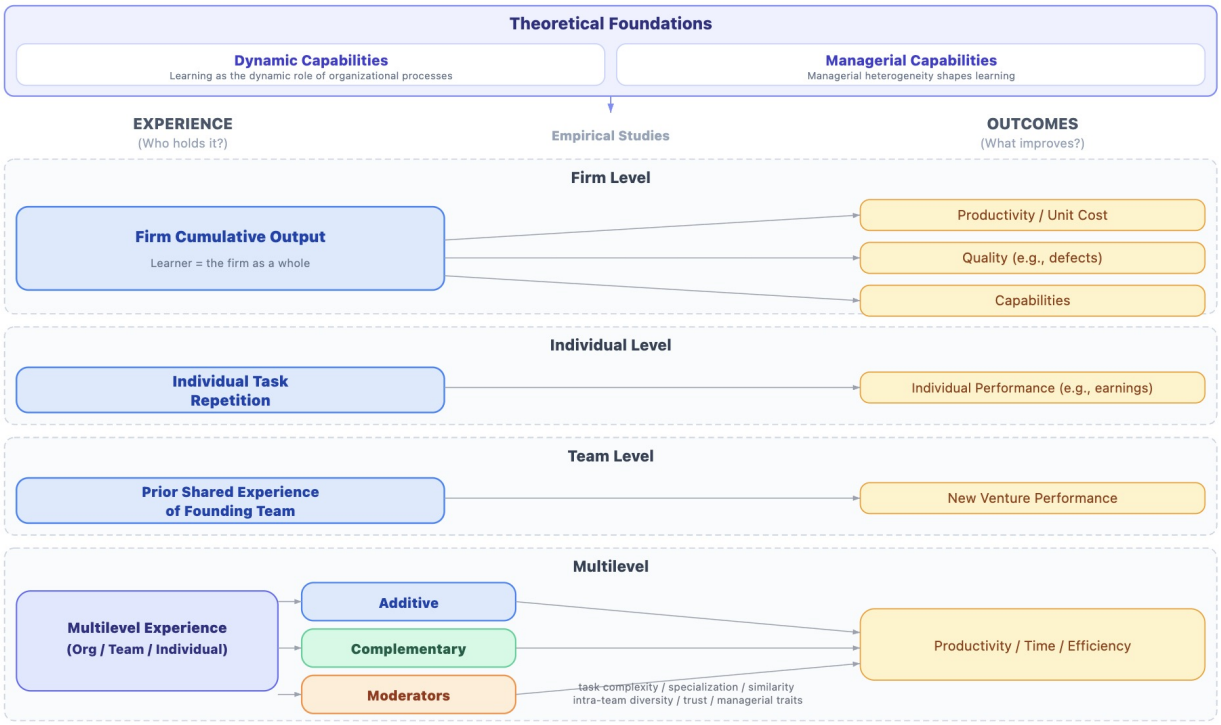


Figure A17: Literature Map

Notes: This figure maps related work on organizational learning along two dimensions. The left side organizes studies by how experience is measured and by the level at which it is assumed to reside (firm, individual, team, and multilevel entities spanning these levels), while the right side groups studies by the performance outcomes examined. For multilevel studies, the figure further distinguishes the main empirical specifications in prior work—additive models combining multiple experience measures, models emphasizing complementarities among experience measures, and models that incorporate moderators—highlighting where the literature clusters and where evidence on particular combinations remains sparse.

Table A19: Summary of Related Literature

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Panel A. Theoretical work							
Harvey et al., 2022 (Academy of Management Annals)	Team (managerial decisions can affect team learning)	-	-	Variation in managerial decisions should produce variation in team learning routines, which then drives capability development and downstream outcomes.	We explicitly test the complementarity between manager learning and worker learning.	"More learning is always better" is an unsafe assumption; learning involves strategic trade-offs. They also call for differentiating team performance measures.	Support our approach to explore the mechanism of team learning and use multiple outcomes.
Helfat and Martin, 2015 (Journal of Management)	Top management teams	-	-	Managers differ in dynamic managerial capabilities rooted in human capital, social capital, and cognition; this heterogeneity can produce differences in strategic change and organizational performance, subject to organizational context.	They synthesize evidence on individual managers and top management teams as sources of strategic change. We examine a different level and process: shared production experience accumulated by exact middle-manager combinations and worker combinations.	Firms should assess and develop the human, social, and cognitive foundations of dynamic managerial capabilities while recognizing that their value depends on context.	Their review supports treating managers as heterogeneous contributors to organizational outcomes. Our analysis extends that insight to operational learning but measures combination-level shared experience rather than classifying individual managers by capability.
Helfat and Peteraf, 2015 (Strategic Management Journal)	Top Managers and the CEO	-	-	Individual managers form the microfoundations of firms' dynamic capabilities. They introduce the concept of "managerial cognitive capability," which highlights the fact that capabilities involve the capacity to perform not only physical but also mental activities.	We focus on middle mangers, instead of top managers. Also, we admit the worker's role in learning, but on different operational outcomes.	Future research should move beyond CEOs to middle managers.	Support our focus on middle managers.
Hodgkinson and Healey, 2011 (Strategic Management Journal)	Managers at all levels	-	-	Organizational learning for dynamic capabilities improves when firms integrate emotion, intuition, and self-regulation into learning routines.	Instead of psychological traits, we focus on manager experience.	The development and maintenance of dynamic capabilities requires firms to harness managers' reflexive and reflective abilities, to utilize implicit and explicit cognitive and emotional processes in harmony, to facilitate sensing, seizing, and reconfiguration.	Dynamic capabilities (learning) reside both in managers and workers.

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Teece et al., 1997 (Strategic Management Journal)	Collectively by the organization	-	-	Learning is a core microfoundation of dynamic capabilities: "Organizational processes have three roles: coordination/integration (a static concept); learning (a dynamic concept); and reconfiguration (a transformational concept)."; Learning is collective, social, and embedded in routines: "Learning requires common codes of communication and coordinated search procedures."	We empirically decompose this collective process into different actors-middle managers and workers.	-	-
Panel B. Level of Analysis: Firm							
Alcacer and Oxley, 2014 (Strategic Management Journal)	The firm as a whole	(1) Suppliers' technological capabilities and (2) marketing capabilities	The duration and extent of supply experience (measured by the number of years and number of customers)	An OEM supplier's accumulation of capabilities will be positively related to the duration and extent of its accumulated supply activity in the industry.	Firm-level analysis	-	-
Anand and Khanna, 2000 (Strategic Management Journal)	Managers responsible for strategic alliances (the implicit learners)	Abnormal Returns	(1) Cumulative number of joint ventures; (2) Cumulative number of licensing agreements	Cumulative joint ventures and cumulative licensing agreements entered into by the firm can increase the firm's excess return.	This paper documents significant learning effects in the management of alliances. This is largely the learning of top managers. However, we focus on middle managers who serve as bridge between top managers and workers.	-	-
Balasubramanian and Lieberman, 2010 (Strategic Management Journal)	The plant as a whole	Value added	Sum of real output	The rate of learning-by-doing, as measured by the slope of the learning curve, will be higher in industries with greater complexity; The heterogeneity of firm performance will be greater in industries with higher rates of learning.	They treat industry learning as a firm-level "black box" where complexity leads to higher learning rates. We open this box by identifying hierarchical loci, showing that learning happens differently for workers and managers based on the specific operational outcome (execution vs. coordination).	Firms in high-learning industries must master the learning process to avoid falling into the "worst-performer" deciles, as these industries exhibit the highest risk and performance heterogeneity.	Identifying that an industry has a "high learning rate" is insufficient for an internal operational strategy. Our results imply that a firm can navigate these high-dispersion environments more effectively by strategically managing hierarchical human capital: applying worker shared experience to execution problems and manager shared experience to coordination bottlenecks.

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Benkard, 2000 (The American Economic Review)	The firm as a whole	Labor hours required per airframe	(1) Production Experience; (2) Industry Experience; (3) General Firm Experience (measured by cumulative output)	Labor hours per unit of aircraft decreases with cumulative past output; there is depreciation of experience (organizational forgetting).	Benkard estimates an aggregate firm-level learning curve. We disaggregate experience by hierarchical role.	To avoid layoffs, even if workers are temporarily underutilized.	Benkard's retention prescription protects the firm's stock of production experience. Our results add that retention and assignment stability may need to be targeted to the role and outcome at issue.
Clark et al., 2018 (Organization Science)	The organization as a whole	Risk-adjusted rework (RARW)	Cumulative experience for hospital	The more experience an organization has with a focal task the better its performance will be with respect to that focal task.	Firm-level analysis	-	-
Egelman et al., 2017 (Management Science)	The firm as a whole	Production volume	Cumulative production volume	Cumulative production volume can increase current production volume, and there is cross-product knowledge spillovers.	Firm-level analysis	-	-
Eggers, 2012 (Strategic Management Journal)	Managers of product portfolio (the implicit learners)	New Product Quality	(1) Experience breadth(measured by concentric diversification index); (2) Experience depth (counts of the firm's funds in the focal new category)	Increases in the focus of a firm's prior product development experiences will result in higher new product quality, but only for products in niches in which the firm has prior experience; and others.	same as Anand & Khanna (2000)	-	-
Epple et al., 1996 (Operations Research)	The manufacturing plant and its production shifts	Shift productivity (vehicle output conditional on direct labor and line hours)	Shift-specific cumulative production and the implied stocks of production knowledge before and after the second shift is introduced	Experience improves plant productivity; knowledge acquired under one-shift operation may transfer to the two-shift regime and across shifts, may accumulate at different rates by shift, and may depreciate over time.	Epple et al. examine the acquisition, retention, and transfer of aggregate production knowledge across shifts. They do not decompose that knowledge into worker- versus manager-combination shared experience; we make that role-based decomposition and relate it to distinct performance outcomes.	When introducing an additional shift, managers should facilitate the transfer of production knowledge and account for how much prior knowledge is retained, transferred, or depreciated.	Their results support actively managing knowledge transfer across organizational units. Our results add that worker- and manager-combination experience are associated with different outcome domains, without attributing shift-level learning solely to workers or to insufficient managerial input.
Hatch and Dyer, 2004 (Strategic Management Journal)	Workers	Defect density per wafer	Cumulative volume; Cumulative engineering	Both cumulative volume and cumulative engineering can reduce defects; Firm-specific human capital is key to improve learning-by-doing.	same as Hatch & Mowery (1998)	To improve the learning curve: Invest in HC selection; Invest in HC development that builds problem-solving capability; Invest in HC deployment.	Human capital include workers and managers. If they have different roles in learning, then there are different ways in human capital selection, training and deployment.

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Hatch and Mowery, 1998 (Management Science)	The manufacturing organization, with engineering problem-solving as a central learning input	Defect density per wafer	Cumulative volume; Cumulative engineering	Both cumulative volume and cumulative engineering can reduce defects.	Their cumulative-volume and engineering-input measures capture plant-level production experience and deliberate engineering problem solving, not shared experience of exact worker combinations. Both studies connect experience to quality improvement, but our estimand distinguishes worker- and manager-combination experience within the same production setting.	Learning is primarily influenced by the systematic allocation of engineering labor to problem-solving activities. In other words, learning is subject to managerial discretion and control.	Their findings support directing scarce engineering resources toward problem solving and knowledge transfer. Our results add a distinct, narrower pattern: worker-combination shared experience is associated with fewer defects, whereas manager-combination shared experience is associated with coordination outcomes.
Hayward, 2002 (Strategic Management Journal)	Managers responsible for acquisitions (the implicit learners)	Acquisition performance	Sum of recent acquisitions	Acquisition experience may be insufficient to ultimately ensure superior acquisition performance.	same as Anand & Khanna (2000)	-	-
Hoang and Rothaermel, 2010 (Strategic Management Journal)	The firm as a whole	(1) Drug approval; (2) termination hazard	External exploration/exploitation experience (percentage of R&D alliances entered into/licensing and manufacturing agreements); Internal exploration/exploitation experience (past R&D efforts and product sales)	External exploitation (exploration) experience has a positive (negative) effect on R&D project performance.	Firm-level analysis	-	-
Kellogg, 2011 (Quarterly Journal of Economics)	The firm as a whole	Drilling rate (feet/day)	Field experience, firm experience, rig firm experience, joint experience (measured by number of wells drilled)	Drilling firm and rig firm's joint experience can increase drilling firm's productivity (relation-specific learning).	Firm-level analysis	-	-

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Kim et al., 2009 (Organization Science)	The firm as a whole	Firm failure rate	Success experience (the discounted sum of the total number of quarters in which the bank recorded a CAMEL rating of 200 or higher since its charter); Recovery experience (the discounted sum of times when a critical financial distress that puts it at high risk of failure overcomes the crisis and returns a normal level of performance)	Both types of experience have an inverted U-shaped relationship with failure rate; the two types interact to enhance each other's value.	Firm-level analysis	-	-
Lafontaine and Shaw, 2016 (Journal of Labor Economics)	Firm owner	Duration of business	Total number of businesses opened before	The number of businesses opened prior to the focal business has a positive effect on its duration.	This paper documents significant learning effects in entrepreneurship. However, we focus on middle managers.	-	-
Levitt et al., 2013 (Journal of Political Economy)	Possibly management	Average defects per car	Cumulative production	Defect rates fall with cumulative production.	Their results indicate that the knowledge obtained through learning by doing is not retained by the workers. It is instead embodied in "physical and organizational capital" via institutionalized systems and managerial practices. In our setting, we find worker experience is a primary driver of quality learning. At the same time, consistent with their emphasis on organization/management for productivity systems, we find manager experience is crucial for coordination outcomes.	Management practices that hasten the learning process (such as creating mechanisms that draw production knowledge out of line workers and quickly embody it in the broader capital of the plant...) result in greater benefits to the firm.	We should first know who possess which kind of knowledge
Pisano et al., 2001 (Management Science)	The organization as a whole	Procedure time (minutes)	Number of prior procedures at hospital	Number of prior cases preformed can decrease procedure completion time for a hospital, and learning rates at different hospitals differ. Therefore, learning-by-doing may be a firm-specific capability.	We show where this capability resides.	Learning from experience is not automatic. Unless an organization puts into place mechanisms for capturing knowledge and implementing learning, experience may not translate into competence.	We go a step forward to show the mechanisms of learning– who improves what outcomes in the firm.

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Salomon and Martin, 2008 (Management Science)	The firm as a whole	Time-to-Build	Firm-level and industry-level plant experience (the number of times that a firm/other firms within the industry has previously built production facilities)	Time-to-build (the time it takes a firm to build and ramp up operations at a new manufacturing facility) will decrease with cumulative firm experience and cumulative industry experience.	Firm-level analysis	-	-
Thornton and Thompson, 2001 (The American Economic Review)	The firm as a whole	Labor requirement per ship	(1) Same design/yard experience; (2) Other designs/yards experience (measured by cumulative output)	Unit labor requirements decrease with cumulative output; there exist cross-design and cross-yard spillovers.	Instead of focusing on transfer of knowledge across models, we show who holds experience matters for model-specific learning	-	-
Weigelt, 2009 (Strategic Management Journal)	Managers involved in the technology adoption and integration process (the implicit learners)	(1) Integrative capabilities; (2) Market performance	Dummy: Prior PC banking	Outsourcing can hurt a firm's subsequent integrative capabilities and performance in the market; and experience in prior related technology may limit the downside of outsourcing.	same as Anand & Khanna (2000)	-	-
Panel C. Level of Analysis: Individual							
Haggag et al., 2017 (AEJ: Applied Economics)	Taxi drivers	Shift productivity (fare earnings)	Shift number; cumulative drop-offs; location-specific drop-offs	Cumulative shifts and drop-offs can increase the fare earnings of taxi drivers.	Haggag et al. estimate learning from an individual driver's own experience. We instead estimate shared experience accumulated by exact worker and manager combinations. Our coefficients therefore describe role-specific combination experience, not the productivity effects of individual workers or managers.	-	-
Ibanez et al., 2018 (Management Science)	Radiologists	Deviation from queue	Radiologist tenure (years)	The probability that an individual deviates from the next task in the queue increases with that individual's level of experience.	same as Haggag et al. (2017)	-	-
Kc et al., 2013 (Management Science)	Surgeons	Patient mortality rates	Cumulative volume of personal successes and failures, and cumulative volume of others' successes and failures	Individuals learn more from their own successes than their failures; they learn more from others' failures than others' successes; prior successes and others' failures help individuals learn from their own failures.	same as Haggag et al. (2017)	-	-

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Lee and Park, 2024 (Strategic Management Journal)	Surgeons	Patient risk-adjusted survival rate (RASR)	Surgeon's own accumulated failures (the cumulative number of patient deaths experienced by a surgeon in a focal hospital)	There will be an inverted-U-shaped relationship between a surgeon's own accumulated failures and learning, such that a surgeon's subsequent surgery performance will improve as a function of their own accumulated failures up to a point but will deteriorate once that point is passed.	same as Haggag et al. (2017)	-	-
Narayanan et al., 2009 (Management Science)	Engineers	Resolution times for maintenance tasks	Specialized experience (number of tasks in a module), task variety (unique modules worked on), and the Herfindahl-Hirschman Experience Index (HHEI) to measure experience concentration balance	Specialization enhances productivity; task variety has a nonlinear, inverted U-shaped effect; and achieving a balance between specialization and variety maximizes productivity.	same as Haggag et al. (2017)	-	-
Peterson and Wu, 2021 (Strategic Management Journal)	Entrepreneurs	Delivery delay (days)	Prior project experience (same subtype)	As entrepreneurs gain experience from past projects, they fail to achieve their predicted delivery date on their next project by a wider margin.	same as Haggag et al. (2017)	-	-
Ramdas et al., 2018 (Management Science)	Surgeons	Surgery duration	Total experience, device-specific experience (first-use markers), and experience gap (time between repeat uses)	The first use of a specific device variant significantly increases surgery duration; knowledge depreciates (forgetting) as the time gap between uses increases.	same as Haggag et al. (2017)	-	-
Panel D. Level of Analysis: Team and Multilevel Analysis							
Adhvaryu et al., 2023 (Review of Economic Studies)	Workers (middle managers can increase their learning rates)	Line efficiency per style	Days spent on same style	Days spent on same style can increase line efficiency. Production lines under different line managers have different learning rates, which is explained by the managers' non-cognitive skills such as autonomy and attention.	We directly interact worker and manager experience to test complementarities-thus focusing on a dynamic, accumulating input rather than a relatively stable managerial trait.	Firms should screen for or train managers in non-cognitive skills like "Internal Locus of Control" and "Attention" to boost worker productivity.	Training and non-cognitive-skill development remain compatible with our implications. Our results add retention policies, assignment stability, and rotation design as complementary levers for building manager shared experience.

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Chen, 2021 (The American Economic Review)	Team	30-day mortality	Shared team experience (the number of hospital visits the physician provided to the proceduralist's patients in the past two years)	Shared work experience of the proceduralist-physician team can improve patient outcome (mortality).	One would expect shared work experience of managers and workers in the same team can improve team outcomes uniformly. However, we show that workers' shared experience improves execution, and managers' shared experience improves coordination.	To re-arrange proceduralist-physician teams to achieve higher shared work experience.	It is important for the firms to know which outcome they are aiming at. If they want to improve execution (coordination), they should prioritize worker (manager) shared experience, rather than just the general team-wide shared experience.
Criaco et al., 2022 (Journal of Management)	Founding team	(1) Time to first foreign market entry; (2) New venture performance: operating profit per employee, and export sales divided by total sales	Prior shared experience (whether they have worked concurrently for a business with foreign market sales anytime between 1997 and the year before starting their new venture for each dyad in the team, averaged across dyads)	(1) Negative relationship between founders' prior shared international experience and time to first foreign market entry. (2) Founders' diversity (industry dynamism) strengthens (weakens) the negative relationship between founders' prior shared international experience and time to first foreign market entry.	same as Kor (2003)	-	-
Espinosa et al., 2007 (Organization Science)	Workers and teams	Length of time to complete an error-free MR project	(1) Task familiarity (the number of prior deltas developed for the subsystem by each team member, averaged across members for the entire team). (2) Team familiarity (the number of prior MRs in which both members had developed code for each dyad in the team, averaged across dyads)	(1) Task familiarity is more beneficial with more complex tasks (i.e., tasks that are larger or with more complex structures). (2) Team familiarity is more beneficial when team coordination is more difficult (i.e., for larger or geographically dispersed teams). (3) The effects of task familiarity and team familiarity on team performance are complementary.	(1) Same as Chen (2021); (2) We test cross-hierarchical complementarity in the same team, and we show this complementarity is outcome-contingent.	They find team familiarity helps teams offset the negative effect of coordination complexity, suggesting firms should make investments to develop team familiarity (e.g., frequent trips to each other's sites, forming teams with members that have worked together, implementing technologies like video conferencing that foster team familiarity).	If a firm's specific problem is coordination, our results imply they should specifically prioritize the shared experience of the managers, rather than the team as a whole.
Fong Boh et al., 2007 (Management Science)	Workers, Teams, Organizational Units (in separate specifications)	Software Development Productivity (labor-hours to complete a unit of work)	Same-system, related, unrelated experience (the number of work units completed); Shared experience at individual, team, unit levels (the number of distinct developers with whom an individual has worked)	Individual/ group/organizational-unit's current productivity in completing work on a system is affected by prior experience from work on same/related/unrelated system, and shared experience with others.	They test experience measured at different levels in three separate regressions. We put manager and worker experience in the same regression-thus doing a within-team decomposition.	To emphasize the learning opportunities from working with others in software development.	Same as Chen (2021)

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Honoré, 2022 (Academy of Management Journal)	Founding team	Startup failure	(1) Number of founders with variety of experiences (in the five years before joining their startups, worked in at least two firms and did not transition from the same industry incumbent as other founding members). (2) Shared experience of founders transitioning from the same industry incumbent (years worked together in the same incumbent firm right before creating the startup)	(1) Number of founders with a variety of firm experiences negatively/positively relates to likelihood of failure. (2) Shared experience of founders transitioning from the same industry incumbent negatively relates to likelihood of failure. (3) Shared experience in the industry incumbent moderates the relationship between variety of experiences and startup failure.	same as Kor (2003)	-	-
Huckman et al., 2009 (Management Science)	Workers and teams	(1) output quality measured by post-delivery defects; (2) adherence to schedule and effort estimates	Team familiarity (previous times members worked together) and role experience (years in current roles, such as manager or engineer)	Team familiarity is positively related to performance; role experience is also positively related to team outcomes.	Same as Chen (2021)	Managers should consider both individual role experience and team familiarity when making staffing decisions.	Same as Chen (2021)
Jain, 2013 (Organization Science)	Worker, team, firm	Innovative productivity (1/ULR)	Individual experience, firm experience (cumulative patents) and team experience (the average number of times pairs of scientists in a team have worked together in the past)	Innovative productivity increases with accumulated individual/team/firm experience; the effect of one kind of experience depends on the other kind of experience.	Jain decomposes innovative experience across individual, team, and firm levels. We instead decompose shared experience within a production unit by hierarchical role and examine whether worker- and manager-combination experience map onto different performance outcomes.	Individual experience is a central repository of innovative capability, while team and firm experience also shape productivity; staffing decisions should therefore consider both domain expertise and collaborative experience.	Our evidence does not establish that either form of shared experience is required. It shows role-aligned associations: worker-combination experience predicts fewer defects, while manager-combination experience predicts shorter waiting times and fewer material shortages. These findings complement, rather than displace, individual-skill considerations.

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Kor, 2003 (Organization Science)	Top management team	Annual rate of sales growth	(1) Founder-based firm-specific experience (ratio of founders active in TMT to TMT size). (2) Team-specific management experience (years of shared experience for the four longest-tenured managers). (3) Industry-specific management experience (average number of prior managerial positions held in same industry)	(1) Each experience type is positively related to entrepreneurial growth. (2) The positive relationship between founder-based firm-specific experience and growth weakens as shared team-specific experience (or industry-specific management experience) increases.	Kor focuses on the Top Management Team (TMT) as a single unit. We show that shared experience is role-specific. Also, instead of TMT, we study middle managers who serve as the bridge between TMT and workers.	Because of conflict effects, the benefit of founder-based experience weakens as either the shared team-specific experience or industry-specific managerial experience increases. She then suggests firms should balance experience levels in the TMT and avoid high levels of shared team-specific/industry-specific experience.	Shared experience is not a monolithic risk. The goal should not be to "avoid" shared experience, but to target it to the hierarchy level corresponding to the desired outcome.
Reagans et al., 2005 (Management Science)	Worker, team, org	Procedure time	Org, individual, and team experience (measured by cumulative number of operations)	Organizational experience, individual experience and team experience reduces procedure completion time.	Same as Chen (2021)	Although each kind of experience represents a distinct kind of knowledge, the different kinds of knowledge can substitute for each other. To compensate for turnover, firms should ensure teams contain at least two experienced members who have worked together before.	The "two members" rule is insufficient because it ignores vertical alignment. Our study suggests that if coordination is the bottleneck, those two experienced members must be managers to effectively improve coordination outcomes.
Schilling et al., 2003 (Management Science)	Team	Game score	Cumulative games played	Cumulative games played improve game score. The learning rate will be greatest under the conditions of specialization/ related variation/ unrelated variation.	They examine how task-content variation shapes team learning. We hold the focal product model constant and distinguish shared experience accumulated by worker and manager combinations.	They find that related task variation improves group learning rates more than specialization, suggesting managers should assign teams to different but related problems over time to foster "learning to learn".	Task variety and shared-combination experience are distinct learning levers. Our results associate manager-combination experience with coordination outcomes and provide suggestive evidence of worker-manager complementarity for flow-sensitive outcomes; they do not show that manager experience directly increases workers' learning rates.
Taylor and Greve, 2006 (Academy of Management Journal)	Team	(1) Mean performance (average collector valuation); (2) performance variance (extreme success or failure)	Genre experience (a count of all genres that the creators on a team had worked with in their careers) and team experience (the number of times a creative team had worked together previously)	Team size and knowledge diversity increase performance variance; team experience increases both variance and mean performance.	same as Chen (2021)	To achieve innovation and high performance, firms should either find "super" individuals or form "fantastic" teams where everyone has deep shared experience.	The "fantastic team" recommendation is too broad. Your study implies that firms should be more surgical: if the goal is to fix execution issues, they need a "fantastic worker team"; if the goal is to fix coordination issues, they need a "fantastic manager team."

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Table A19 continued

Paper	Who the Learners Are	Dependent Variable	Measures of Experience	Theoretical predictions	Contrast to our results	Their implications	Contrast to (or support for) our implications
Zheng, 2012 (Journal of Business Venturing)	Founding team	New venture growth (composite of self-reported sales, employee, and market share growth)	Prior shared experience (worked together directly for at least one year in prior organizations for each dyad in the team, averaged across dyads)	(1) Impact of prior shared experience on new venture performance is partially mediated by a founding team's TMS. (2) Positive relationship between TMS and performance becomes stronger as task similarity (and intra-team trust) increases.	He identifies the TMS as the team-level cognitive mechanism that makes PSE effective for growth. We provide a more granular structural view: instead of a general team-level TMS, we show that the locus of learning is divided hierarchically. Worker shared experience specifically improves execution, a nuance not captured by general team-level cognition models.	Founding teams should leverage TMS to coordinate expertise; teams lacking PSE should use group training to nurture these cognitive structures.	General "group training" to build TMS is insufficient because it treats all team members as equal learners. Our results imply that firms should bifurcate training efforts: focus on building shared experience among workers for execution-heavy tasks and among managers for coordination-heavy tasks.
Zheng et al., 2016 (Strategic Management Journal)	Founding team	Return on assets (ROA)	(1) PSE extensiveness (length of overlap in career histories in same prior banking institution for each dyad, averaged across dyads). (2) Current shared experience (months the team operated during its initial year)	(1) New firms with more extensive PSE outperform those with less extensive PSE. (2) Relationship between PSE extensiveness and performance diminishes as a founding team accumulates shared operating experience / as PSE task environment mismatch increases / as PSE temporal gap increases.	Zheng et al. examine how current shared history and contextual fit condition the value of prior shared experience. We distinguish current shared experience by hierarchy: worker-combination experience is associated with fewer defects, manager-combination experience is associated with coordination outcomes.	Investors and regulators should prioritize teams whose PSE context (e.g., bank charter type) matches the new venture's environment, as mismatched experience provides no benefit over having no shared experience at all.	Context matching and within-role shared experience are complementary considerations. Our evidence supports building worker shared experience when defect reduction is the objective and manager shared experience when coordination is the objective.
Chen et al. (2026)	Workers and middle managers	Execution outcomes (Operating Time and Defects); Coordination outcomes (Waiting Time and Material Shortages); Productivity	Worker shared experience and manager shared experience measured by cumulative production	H1: Workers' shared experience primarily improves execution performance, whereas managers' shared experience primarily improves coordination performance. H2: Greater managerial interventions accelerates both managers' and workers' learning. H3: Worker and manager experience are complementary, such that each strengthens the other's performance benefits in its primary domain. H4: Managerial soft skills strengthen worker–manager complementarity for coordination outcomes, while their moderating effect for execution outcomes is theoretically indeterminate.	-	-	-

Appendix References

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